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Twelve millennia of plant use in the Northwest Amazon: archaeobotanical insights from Cerro Montoya 1

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Cerro Montoya 1 (CM1), in the Serranía de La Lindosa (Guaviare, Colombia), preserves one of the longest and most detailed multi-proxy archaeobotanical sequences in north-western Amazonia, spanning from the Late Pleistocene (~12.6 cal ka BP) to the early Colonial period (~0.4 cal ka BP). Integrated analyses of plant macroremains, phytoliths, starch grains, and charred parenchyma reveal a remarkably stable, broad-spectrum subsistence strategy from the earliest occupations onward. Palms—especially *Syagrus orinocensis*, *Attalea maripa*, and *Astrocaryum chambira*—dominate the assemblage throughout the sequence, underscoring their central dietary and technological importance. This study also provides one of the first systematic archaeobotanical documentations of root and rhizome exploitation in the Colombian Amazon, identifying geophytes from Costaceae, Marantaceae, Heliconiaceae, and Strelitziaceae through microfossil and anatomical evidence. CM1 also yields the earliest evidence in north-western Amazonia for the use of Brazil nuts (*Bertholletia excelsa*, ~7.2–7.0 cal ka BP). Increasing palm abundance and diversity through the Holocene suggest low-level, cumulative forms of forest management rather than early domestication. Cultivated crops are absent during the initial millennia of occupation and appear only in the Late Holocene, with maize (*Zea mays*) restricted to the final pre-Columbian layers (~0.5–0.3 cal ka BP). The assemblage reflects strategic exploitation of a heterogeneous ecotonal landscape incorporating *terra firme* forest, flooded forest, savannah, and tepui mesatop hill environments. Combined with faunal evidence, the data indicate that early inhabitants entered this transitional Amazon–Orinoco region with fully developed, flexible subsistence systems centred on palms, roots, and forest fruits, rather than on incipient cultivation. CM1 thus contributes critical evidence to debates on tropical forest colonisation, early plant management, and the long-term resilience of foraging economies in Amazonia.

KEYWORDS
Amazonia, early human dispersal, geophytes, macro-botanical remains, neotropics, palms, phytoliths, starch grains

1 Introduction

The plant component of early South American diets remains one of the least explored aspects of the initial human dispersals into the continent. This reflects a combination of factors: early sites are rare, there is limited research coverage, the preservation of macrobotanical remains is generally poor, and systematic flotation or microbotanical analyses have seldom been applied. As a result, the role of plants in early subsistence systems has long been understudied. Yet, a growing body of archaeobotanical research highlights the centrality of plants in early human dispersal and adaptation globally (e.g., Florin et al., 2020), and particularly across South America (Aceituno and Loaiza, 2018; Dillehay and Rossen, 2002; Gnecco and Mora, 1997; Iriarte, 2024; Maggard, 2015; Mora, 2003; Morcote-Ríos et al., 2014; Roosevelt et al., 1996; Shock and Moraes, 2019). Ethnographic and ethnoarchaeological studies further underscore the importance of plant foods in the diets of today's Amazonian hunter-gatherers (Politis and Loperfido, 2025), while recent isotopic and archaeobotanical evidence indicates that in several regions, plants may have been an important dietary component of the first Americans, challenging long-standing assumptions that early subsistence economies were primarily based on the hunting of large mammals (e.g., Chen et al., 2024).

The Serranía de La Lindosa (SLL) in the Colombian Amazon offers a unique early opportunity to address this imbalance. Situated at the ecological crossroads between the Orinoco Llanos, savannahs and the Amazon rainforest, its sandstone rockshelters preserve one of the richest Late Pleistocene-Early Holocene archaeological landscapes in the Neotropics, combining twelve thousand year stratigraphic sequences with exceptional preservation of charred organic remains, lithic assemblages and rock art (Aceituno et al., 2024b; Iriarte, 2024; Iriarte et al., 2022a; Morcote-Ríos et al., 2021). Within the SLL, the Cerro Montoya 1 rockshelter (CM1) provides an unparalleled record for a diachronic examination of plant use since the earliest human occupations in north-western Amazonia.

In this paper, we present a multiproxy archaeobotanical study of CM1, integrating analyses of macrobotanical remains, phytoliths, and starch grains. The site contains well-preserved deposits dating back to at least 11.8 cal ka BP, which reveal the plants consumed and utilised since the region's initial human settlement. The results show that palms (particularly *Syagrus orinocensis*, *Attalea maripa*, and *Astrocaryum chambira*) dominate the assemblage from the earliest levels of human occupation. Other taxa include fruit trees (*Brosimum lactescens* (Moraceae), *Spondias mombin* (Anacardiaceae), geophytes (Costaceae, Marantaceae, Heliconiaceae, and Strelitziaceae (*Phenakospermum guyannense*), and resin-producing trees [*Protium* sp. (Burseraceae), *Hymenaea courbaril* (Fabaceae)]. Taken together, these data indicate a broad-spectrum subsistence strategy focused on year-round plant resources drawn from a mosaic of flooded forests, terra firme, and savannah margins. Throughout the Holocene, the sustained abundance and taxonomic diversity of palm remains suggest low-level management practices, including selective harvesting and the dispersal of seeds around habitation areas.

The remarkable preservation of botanical remains also allows insights into plant management and medicinal use. Several taxa, including *Protium* sp. (Burseraceae), *Costus spiralis (Zingiberales)*, *Phytolacca rivinoides* (Phytolaccaceae), and *Byrsonima spicata* (Malpighiaceae), indicate an early Amazonian pharmacopoeia, in which plants served not only as food but also as sources of resins, gums, medicines, and stimulants. Notably, CM1 provides the earliest evidence in north-western Amazonia for the exploitation of Brazil nuts [*Bertholletia excelsa* (Lecythidaceae), ~7,200–7,000 cal ka BP] and for the use of coca leaves (*Erythroxylum coca*, Erythroxylaceae) during the Late Holocene.

2 Materials and methods

2.1 Materials

2.1.1 Study area: Serranía de La Linsosa

The SLL is a 20 km^2 rocky outcrop located in the Department of Guaviare, in the northwest of the Colombian Amazon (Figure 1). The SLL is situated on the banks of the Guayabero/Guaviare River, in today's transitional ecotone between the savannahs of the Orinoco and the Amazon rainforest (Vriesendorp et al., 2018). Geologically, SLL is located on the edge of the sedimentary basins of Vaupés-Amazon to the south and the Eastern Plains (Llanos Orientales) to the north. SLL outcrops are composed of Cretaceous sedimentary rocks of the Araracuara formation. Quaternary rocks and colluvial deposits are found in the alluvial valleys and SLL foothills. The soils can reach a thickness of up to 2 m in the valleys, although they are far shallower in the upland areas.

The climate is warm and humid, and SLL receives ~2,800 mm of rain annually, with a dry season from November to February and a wet season from March to October. The region experiences an average annual temperature of around 28°C, and it is classified within the Tropical Rainforest life zone (Cárdenas et al., 2008). A diversity of habitats characterises SLL owing to its being in an ecotonal zone that straddles wet tropical forests and savannah environments.

Plant inventories of the area have recorded 884 species of vascular plants, corresponding to trees, shrubs, vines, herbaceous plants, and palms, although the total number is estimated to be higher. In areas where the rocky substrate is closer to the surface, the size of trees and shrubs is smaller (Cárdenas et al., 2008; Vriesendorp et al., 2018). Similarly, vertebrate diversity is high, with 449 species, including fish (89), amphibians (30), reptiles (56), birds (226), and mammals (48) (Vriesendorp et al., 2018).

Around the study site, a rapid floristic survey conducted by botanist Hoyos-Gómez (2022) documents a well-developed *terra firme* forest distinct from the vegetation observed on the higher rocky sandstone escarpments of the Cerro Pinturas tepui. CM1 is situated on low-gradient colluvial plains, where deeper soils and more stable moisture regimes support a closed-canopy forest with a diverse arboreal and understorey flora. The vegetation is characterised by a high abundance of palms, including *Oenocarpus bataua*, *O. minor*, *Attalea insignis*, *Astrocaryum chambira*, and the understorey palms *Geonoma deversa* and *G. interrupta*, the latter absent from the exposed rockshelters. Arboreal taxa such as *Iryanthera hostmannii* and *Virola elongata* (Myristicaceae) are particularly frequent, alongside fruit-bearing trees including *Theobroma obovatum* (Malvaceae), *Herrania* sp. (Malvaceae), and

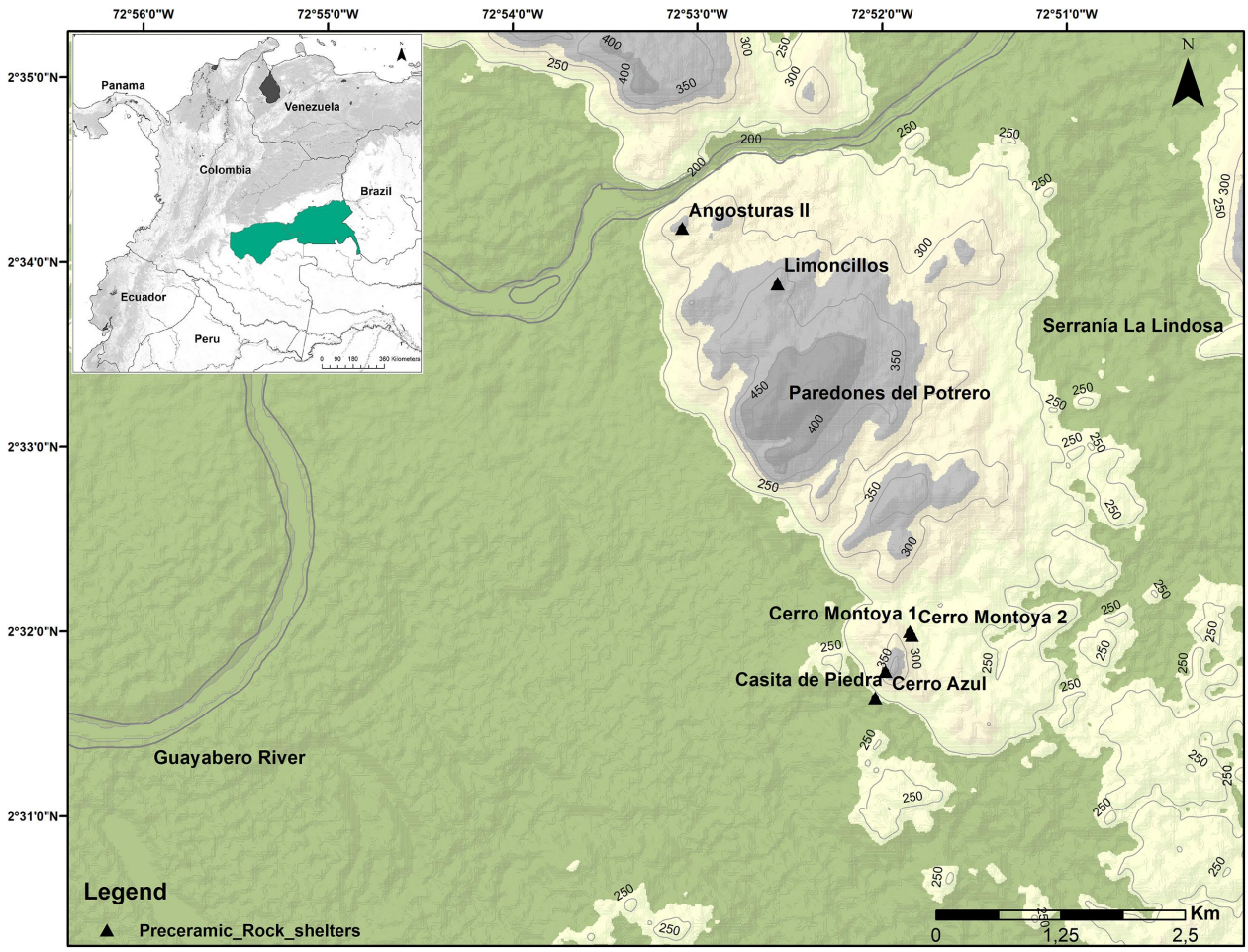


FIGURE 1
Study area map showing location of the Cerro Montoya 1 site and other identified preceramic rockshelters in the Serranía de La Linsosa, Guaviare department, Colombia.

Hymenaea oblongifolia (Fabaceae), all of which are relevant to both past human subsistence and forest fauna. The understorey is rich in Marantaceae (*Ischnosiphon sp.*, *Calathea* sp.), Poaceae (*Olyra latifolia*), Rubiaceae (*Palicourea* sp., *Psychotria* sp.), and Melastomataceae (*Tococa* sp., *Miconia* sp.), reflecting a structurally complex forest floor with high resource diversity. Several of these taxa are edible, medicinal, or have documented technological uses among Amazonian Indigenous groups, and they are present in the archaeobotanical record presented below, underscoring the relevance of the local vegetation mosaic to early human occupation. Overall, the CM1 vegetation represents a resource-rich forest environment that contrasts sharply with the more lithophytic and savannah-associated plant communities elsewhere in the SLL, providing an ecological setting conducive to repeated occupation and intensive plant use (Supplementary Figure SI7).

2.1.2 Archaeological context: the Cerro Montoya 1 rockshelter

To date, six sites with documented preceramic occupations have been identified in the region: Cerro Azul, Cerro Montoya 1 (the focus of this study), Cerro Montoya 2, Limoncillos, Angosturas II, and Casita de Piedra (Figure 1). All are multi-component rockshelters preserving stratified sequences that include both preceramic and ceramic deposits. Radiocarbon evidence indicates a nearly continuous sequence of human occupation in the SLL spanning over twelve thousand years, from the Younger Dryas (~12.6 cal ka BP) to the 16th Century AD, interrupted only by a chronological gap at the end of the Middle Holocene, between approximately 6.0 and 4.0 ka cal ka BP (see details in Aceituno et al., 2024a, 2024b).

CM1 (~273 m a.s.l.) is an isolated sandstone rockshelter located at the base of the Cerro Pinturas tepui. In 2021, two excavation units were opened to compare stratigraphy and activity areas: Unit 2 (3 × 3 m) beneath the overhang and Unit 3 (2 × 2 m) outside the drip line (Figure 2). The stratigraphic deposits, primarily composed of sands eroded from nearby outcrops, contain both preceramic and ceramic occupations separated by clear stratigraphic discontinuities, suggesting episodic rather than continuous use (Aceituno et al., 2024b). Eight archaeological stratigraphic layers were identified in both units, with preceramic occupations in the lower levels (U2: layers IV–V, 180–126 cm; U3: layers II–V, 250–150 cm) and ceramic layers in the upper ones (layers VI–VIII) (Supplementary Figures SI1, SI2).

Radiocarbon evidence indicates that Cerro Montoya 1 (CM1) was occupied intermittently for over twelve millennia, with individual excavation units preserving distinct and discontinuous occupational histories. The earliest evidence for human presence is an isolated

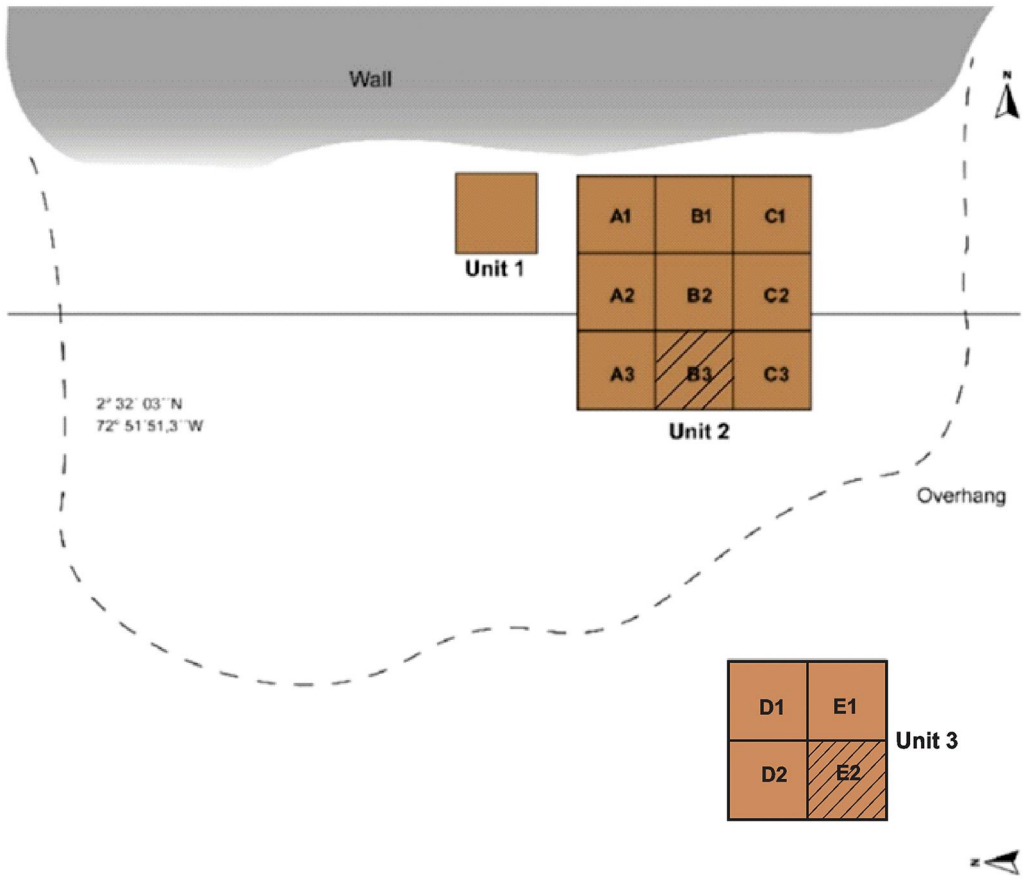


FIGURE 2
Provenience map of the Cerro Montoya 1 site showing the location of selected sediment flotation sampling grids.

date of c. 12.6 cal ka BP from Unit 1 (2018), while the first sustained and stratigraphically coherent occupations occur at the onset of the Holocene (c. 11.8–9.9 cal ka BP). These Early Holocene occupations are well represented in Units 2 and 3 (excavated in 2021), corresponding to Layer IV in Unit 2 (11.7–9.9 cal ka BP) and Layer II in Unit 3 (11.2–10.6 cal ka BP) (Supplementary Figure SI3). Middle Holocene activity is unevenly preserved, occurring in Unit 2 within Layer V (c. 8.3–7.6 cal ka BP) and more continuously in Unit 3 across layers III–V (c. 8.3–6.1 cal ka BP). Notably, substantial chronological gaps are present within individual units, including a pronounced hiatus in Unit 2 between c.

7.5 and 1.9 cal ka BP. Late Holocene occupations, marked by the appearance of ceramics, occur above these gaps in both units (Layers VI–VIII) (Supplementary Figures SI1, SI2). Comparable Early–Middle Holocene hiatuses, lasting several millennia, have been documented at other SLL rock shelters (e.g., Limoncillos and Cerro Azul) and are best interpreted as periods of site abandonment rather than sampling or preservation bias (Aceituno et al., 2024b). For this reason, the Results are presented by excavation unit rather than as a single continuous sequence.

Throughout the sequence, lithic artefacts are predominantly manufactured from local chert (c. 80%), sourced from the beaches of the Guayabero River, with the remainder made of milky quartz. The assemblage comprises flakes, unretouched and retouched tools, cores, and nodules, with most artefacts (86.4%, $n = 5{,}157$) recovered from Unit 2. Technologically, the assemblage is characterised by unifacial reduction strategies that produce short flakes with marginal edge retouch, and it lacks bifacially reduced forms, such as projectile points or knives. Most artefacts are microlithic, measuring under 6 cm, a pattern that reflects the small size of locally available raw materials from the Guayabero River (Aguirre, 2025).

A total of 150 ceramic sherds and one complete vessel were recovered from Unit 2. The assemblage is dominated by coarse, thin-walled, undecorated utilitarian wares, with approximately 90% of the sherds tempered with vegetal material, including charred wood and palm fruit fragments. A small number of decorated sherds can be attributed to the Guayape tradition, characterised by denser fabrics and red-and-cream polychrome decoration, and dated to the Late Holocene (Aceituno et al., 2024b).

2.1.3 Plant processing tools

For the present study, the recovery of artefacts associated with plant processing is particularly significant. The earliest examples were found in contexts dated to 11.3 cal ka BP. In total, approximately 50 implements related to plant processing were recovered—36 from Unit 2 and 14 from Unit 3. Grinding stones (*manos*) occur throughout the sequence and are the only such tools identified in Early and Middle Holocene contexts ($n = 10$). In contrast, ceramic-phase occupations exhibit higher tool frequency and greater technological diversity, including 33 *manos*, six milling stones, and one anvil (Aguirre, 2025).

2.2 Methods

2.2.1 Macrobotanical remains

2.2.1.1 Field

To assess spatial and diachronic variability in macrobotanical plant remains at CM1, we subsampled Units 2 and 3 by designating a 1 × 1 m grid in each unit for focused macrobotanical analysis: Grid B3 (Unit 2) and Grid E2 (Unit 3) (Figure 2). Carbonised plant remains from the selected grids were recovered through a combination of hand collection during excavation, dry sieving, and flotation. For each excavation context —that is, the smallest stratigraphic unit recorded separately in the field, representing a single depositional, cut, constructional, or use event (e.g., Roskams, 2001)— 20 L of sediment were collected and processed using a flotation machine fitted with a 0.5 mm mesh, with additional 0.5 mm mesh cloth bags used to retain small faunal and floral remains (Supplementary Figure SI4).

In total, 34 samples were floated from Grid B3 and 49 from Grid E2, yielding 83 flotation samples and a total of 1,660 L of processed sediment. Following flotation, both light and heavy fractions were submerged in running water for several hours to facilitate cleaning and separation of residual sediment prior to laboratory analysis. All samples were subsequently air-dried in the shade to ensure preservation and safe transport to the laboratory. The application of systematic flotation under tropical rainforest conditions resulted in the recovery of thousands of botanical remains, including fruit fragments, palm endocarps, seeds, plant resins, and charcoal described below.

The selected assemblages show no significant taphonomic disturbances. All remains derive from well-defined archaeological contexts associated with features such as hearths, are consistently charred, and show no evidence of rodent gnawing (Supplementary Figure SI3), root etching, weathering, or post-depositional abrasion. Together, these characteristics indicate good contextual integrity and reliable preservation of the archaeobotanical record.

2.2.1.2 Laboratory

In the laboratory, each sample was sieved through meshes with apertures of 0.5 mm, 2 mm, 4 mm, and 8 mm to facilitate the separation of remains by size and category (seeds, charcoal, resins, microflakes, pigments, and ceramic fragments). This separation step was carried out under a stereomicroscope at 40× magnification and, where appropriate, by direct visual inspection. Once fruit, seed, and plant resin remains were separated, they were quantified, morphologically described, and botanically identified (Supplementary Figure SI4).

The identification of charred plant remains was based on comparisons with the carpotheque of the Archaeology Unit and reference collections of the Colombian National Herbario Nacional Colombiano (COL), National University of Colombia (Supplementary Figure SI6), supported by the local florula of the SLL (Cárdenas et al., 2008) and regional catalogues for Amazonia and the Neotropics (e.g., Lentz and Dickau, 2005). Data analysis was conducted using relative plant counts (presence counts of botanical remains in the selected grids and levels) (Table 1), and graphical outputs were generated using Tilia software (v. 2.1.1).

2.2.2 Phytoliths

2.2.2.1 Field and laboratory

For phytolith analysis, 100 ml of sediment was sampled at 5 cm intervals from the profile wall of Unit 3, Grid E2, resulting in a total of 49 samples analysed.

Phytolith extraction from archaeological sediments at CM1 followed the protocol outlined by Piperno (2006). Phytoliths were examined under a Nikon Eclipse E400 microscope, with photomicrographs taken using a Leica Flexacam C3 camera. For each slide corresponding to the ≤50 µm fraction, 200 phytoliths were counted, while slides containing the ≥50 µm to ≤250 µm fractions were systematically reviewed to identify large-sized phytolith structures.

Phytolith identification was supported by the modern phytolith reference collection of the Institute of Natural Sciences, National University of Colombia, complemented by phytoliths extracted from specimens housed in the Colombian National Herbarium. These included taxa from the order Zingiberales—Cannaceae, Costaceae, Heliconiaceae, Marantaceae, Strelitziaceae, and Zingiberaceae—that occur in the Colombian Amazon and are of significant ecological and cultural relevance (Supplementary Figure SI14 and Supplementary Tables SI1, SI2). Phytoliths were described following the criteria established by the ICPN 2.0 nomenclature (Neumann et al., 2019), with the morphological identification of Zingiberales and palms based on Chen and Smith (2013) and Morcote-Ríos et al. (2016). To further broaden the reference collection and refine phytolith determinations for plants associated with the site, botanical specimens were collected across different biomes of the SLL. Leaf tissue samples of "sweet yuca" and "bitter yuca" (*M. esculenta*) were collected from two Amazonian chagras (cultivated fields) in order to characterise diagnostic heart-shaped phytoliths and to compare them with potential opal morphotypes identified in the CM1 assemblage (Supplementary Figure SI9 and Supplementary Table SI3).

2.2.3 Starch grains

2.2.3.1 Field and laboratory procedures

Starch microfossil analysis was conducted to identify plant taxa exploited at the site. Artefact selection for starch analysis began during excavation with the identification of artefacts potentially associated with the processing of plant resources, such as fruits, rhizomes, or palm hearts. Artefacts selected for starch recovery were immediately wrapped in aluminium foil and stored in airtight plastic bags to minimise the risk of contamination from modern materials.

In total, 16 samples were analysed—seven from ceramic occupations and nine from preceramic occupations—of which two yielded no starch remains. Positive results were obtained from two milling stones, one unifacial flake, one ceramic vessel, and ten handstones (see Table 2).

Sediment adhering to these artefact surfaces was removed using sterile toothbrushes. Starch grains were then extracted from the recovered sediments following established procedures. A heavy liquid solution prepared with caesium chloride (CsCl) at a density of 1.8 g/ml was added to separate starch grains by flotation (Dickau, 2005; Piperno, 2006; Rumold and Aldenderfer, 2016). Samples were centrifuged for five minutes at 2000rpm to allow heavier particles to settle while starch grains remained in suspension. Two millilitres were then removed from the floating fraction and transferred to a

TABLE 1 List of identified plant taxa macroremains from Cerro Montoya 1, Unit 2/ B3 and Unit 3/E2.

Taxon	Unit 2/B3 (Counts)	Unit 3/E2 (Counts)	Total counts
Annonaceae	7	1	8
Xylariaceae (Fungi)	0	12	12
Astrocaryum chambira (Palmae)	1,738	893	2,631
Attalea insignis (Palmae)	1,885	1,497	3,382
Attalea maripa (Palmae)	2,128	877	3,005
Bactris sp. (Palmae)	287	62	349
Brosimum lactescens (Moraceae)	31	40	71
Byrsonima cf. *spicata* (Malpighiaceae)	25	6	31
Cochlospermum sp. (Cochlospermaceae)	0	1	1
Cocoseae *(Ast./Att./Bac./Sya.)* (Palmae)	18,828	10,369	29,197
Costus cf. *spiralis* (Costaceae)	0	12	12
Coussapoa sp. (Urticaceae)	42	18	60
Croton sp. *1* (Euphorbiaceae)	1	0	1
Euphorbiaceae	24	6	30
Euterpe precatoria (Palmae)	11	3	14
Guatteria sp. *1* (Annonaceae)	12	5	17
Guatteria sp. 2 (Annonaceae)	1	0	1
Heliconia cf. *hirsuta* (Heliconiaceae)	0	2	2
Humiraceae	0	2	2
Hymenaea courbaril (Fabaceae)	5	0	5
Leguminoseae	2	17	19
Mauritia flexuosa/Mauritia carana (Palmae)	116	43	159
Morfotipo	418	857	1,275
Oenocarpus bataua (Palmae)	15	7	22
Oenocarpus minor (Palmae)	24	52	76
Pera arborea (Peraceae)	47	26	73
Phytolacca rivinoides (Phytolaccaceae)	0	2	2
Protium sp. (Burseraceae)	3	9	12
Sacoglottis mattogrossensis (Humiriaceae)	21	10	31
Sapium sp. (Euphorbiaceae)	3	5	8
Siparuna sp. (Siparunaceae)	0	5	5
Solanaceae indet.	0	2	2
Spondias mombin (Anacardiaceae)	0	1	1
Syagrus orinocensis (Palmae)	6,805	5,185	11,990
Turpinia sp. (Staphyleaceae)	0	5	5
Zea mays (Poaceae)	5	0	5

clean test tube. The final residue was mounted on microscope slides and examined under a Zeiss AX 10 microscope at magnifications between 200× and 400×. Slides were scanned using a zigzag pattern, beginning in the bottom-right corner.

Archaeological starch grains were described following the International Code for Starch Nomenclature (ICSN, 2011), which is based on the description of metric and morphological variables used for taxonomic identification through comparison with reference specimens (Dickau, 2005; Loy, 1994; Piperno, 2006; Torrence and Barton, 2016). Taxonomic identification was undertaken through comparison with modern specimens from a reference collection of useful and wild plants from the SLL. In addition, numerous specialised publications, particularly those by authors working extensively in the Neotropics (e.g., Ciofalo et al., 2018; Dickau, 2005; Dickau et al., 2007; Jiménez, 2015; Pagán-Jiménez et al., 2015; Perry, 2002; Piperno and Dillehay, 2008) were consulted, along with the reference collection of the Archaeology Laboratory at the University of Antioquia. To monitor potential transfer or contamination, sediment samples collected adjacent to the analysed lithic tools were processed following the same protocol.

TABLE 2 Presence and frequency of starch types by artefact, including handstones (HS), milling stones (MS), and a ceramic vessel, alongside associated lot numbers, stratigraphic context, and calibrated radiocarbon ages (cal ka BP).

Unit	Tool	Level (cm)/C^{14} Cal BP	Fabaceae	*Hymenaea courbaril*	Polyhedral	*Small Bell-shaped*	Small Polyhedral	*Heliconia*	*Costus spiralis*	*Monotagma laxum*	*P. guyanensis*	*Marante arundinacea*	U.S.O	*Teobhroma. obovatu*	*Zea mays*	*Manihot esculenta*
2	Ceramic vessel	5 (45–53) 390	2									3		3	19	
2	HS L.210414	15 (110 −112) 1.8 ka	2					1			2			2	3	1
2	HS L.210412	15, 1.8 ka													2	
2	MS L.210258	13 (93–105) 1.8/1.5 ka	1	Cluster												
3	MS L. 212876	19 (124–127) 2.7/1.7 ka				3							1		2	
3	HS L.212897	20 (127–131) 2.5/1.8 ka	2			2		1	1				2	4	8	
2	HS L210536	17 (120–122) 8.2/1.8 ka	1			1		1	1							
3	HS L214068 150–153	26 (150–153) 6.9 ka	1			3										
3	HS L213165	26 6.9 ka			9	1	3									
3	MS L213415	33 (178–183) 7.9 ka	2		8		3					1		2		
2	HS L212018	21 (132–134) 8.2 ka	2		7			1					4			
2	HS L211368	24 (141–144) 10.2 ka	2		3			3	3	3						
2	HS L211828	26/27 (148–161) 11.2 ka	9		12			2			1	2		2		
2	HS L. 213218	28/29 (162–171) 11.3	1		9	4		1				2				
Total			**24**			**16**	**6**	**23**	**4**	**11**	**4**	**8**	**4**	**17**	**34**	**1**

Bold values in the final row indicate the total number of starch grains assigned to each taxon or morphotype across all analysed artefacts. Blank cells indicate absence or non-identification of that starch type in the corresponding artefact.

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3 Results

Across the macrobotanical and phytolith datasets, we identified 33 taxa, ranging from tribe and order to family, genus, and species level: one at the tribal level (Cocoseae); one order (Zingiberales); four families (Annonaceae, Cyperaceae, Solanaceae, Poaceae); seven genera (*Bactris* Palmae; *Geonoma* Palmae; *Protium* Burseraceae; *Sapium* Euphorbiaceae; *Turpinia* Staphyleaceae; *Licania* (Chrysobalanaceae); *Annulohypoxylon* (Fungi), and twenty species. These findings expand our understanding of past human–ecosystem interactions, including diet, forest cover, catchment areas, and human impacts on the landscape.

In the case of starch remains, it is not possible to determine an exact number of taxa, as some were grouped using criteria that do not strictly follow formal botanical classifications. Moreover, as presented below, certain morphotypes may correspond to more than one taxon. The most securely identified taxa include starches attributable to the Fabaceae family; three rhizomatous plants (*Costus cf. spiralis* Costaceae; *Heliconia acuminata* Heliconiaceae; and *Heliconia cf. psittacorum* Heliconiaceae); and a group of starches classified as Underground Storage Organs (USOs), encompassing *Dioscorea* spp. (Dioscoreaceae) and *Phenakospermum guyannense* (Strelitziaceae). Within the Marantaceae family, starches consistent with *Maranta arundinacea* (arrowroot) and *Monotagma cf. laxum* (wild rhizome) were identified. In addition, starches attributable to *Theobroma cf. obovatum* (Malvaceae) wild cacao, *Zea mays* (Poaceae) maize, and *Manihot esculenta* (Euphorbiaceae) manioc, were recorded.

In the following sections, we present the results organised by proxy: macrobotanical remains, phytoliths, and starch grains.

3.1 Macrobotanical remains

We recovered abundant carbonised plant remains from the sampled grids in both units. Below, we summarise overall quantities and then describe the assemblage chronologically by unit. Unit 2, Grid B3 yielded 32,484 specimens. Unit 3, Grid E2 yielded 20,032 remains. The total assemblage from both grid squares amounted to 52,516 remains, from which 36 taxa were identified: 19 species, 12 genera, 4 families, and one tribe (Table 1). In the following section, macrobotanical remains are presented in chronological order to illustrate temporal changes in taxonomic composition, patterns of predominance, and shifts in plant use.

Quantitative comparisons based on seed counts should be interpreted with caution. Taxa differ markedly in preservation potential and reproductive output, and these factors may bias observed abundances. In particular, palms produce large quantities of seeds and possess thick, robust endocarps that are more resistant to charring and post-depositional processes than the smaller, thinner-walled seeds of many fruit trees. As a result, palm taxa are more likely to be preserved and recovered archaeologically and may be over-represented relative to their actual importance in past plant use.

3.1.1 Cerro Montoya 1. Unit 2, Grid B3

This section presents the stratigraphic distribution and composition of plant macroremains recovered from Unit 2, Grid B3 at CM1, documenting changes in plant use from the Early Holocene to the Late Holocene (Figure 3).

3.1.1.1 Early Holocene: layer IV (11.7–10.1 cal ka BP)

A total of 2,377 archaeological seeds were recovered for this period. Thirteen plant taxa were identified, with a strong predominance of palms, including *Syagrus orinocensis*, *Astrocaryum chambira*, *Attalea maripa*, and *A. insignis*. Endocarp remains identified at the tribal level (Cocoseae) are particularly prominent in the archaeological record for this period. Other palms, such as *Mauritia flexuosa/M. carana*, *Bactris* sp., *Oenocarpus minor*, and *O. bataua*, occur in low frequencies.

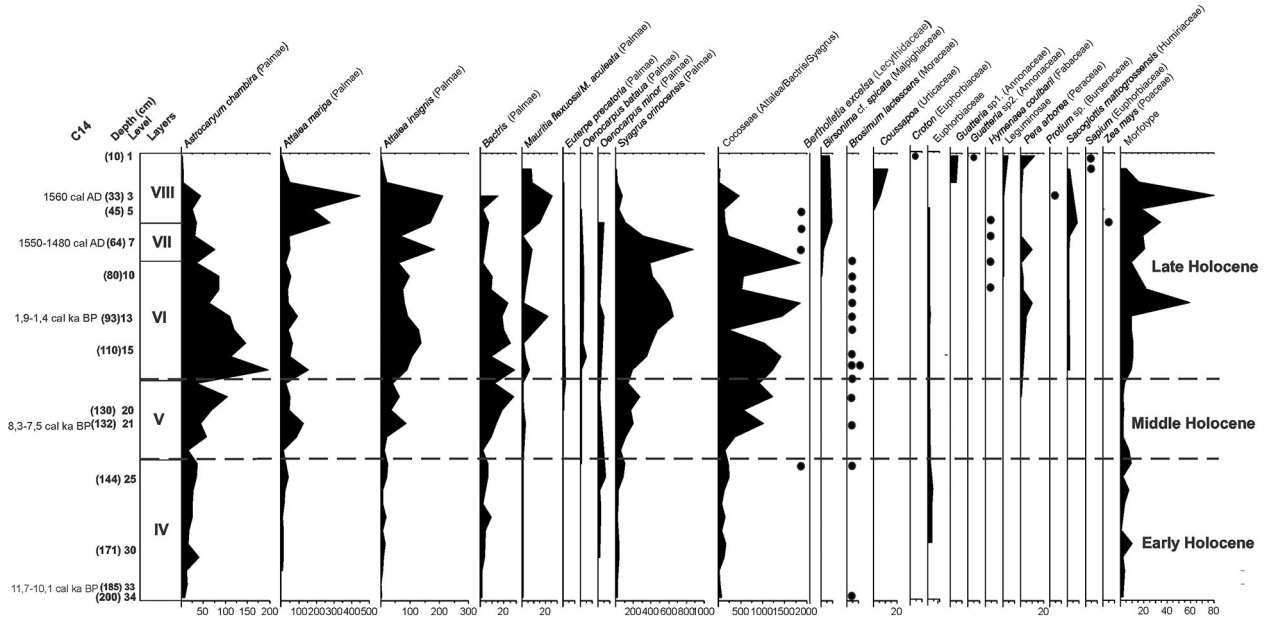


FIGURE 3
Stratigraphic diagram of seed counts from the flotation of Cerro Montoya 1 Unit 2, Grid B3.

Three arboreal taxa show limited representation during this period: *Sapium* sp. Euphorbiaceae, *Brosimum lactescens* (Moraceae), and *Bertholletia excelsa* (Lecythidaceae), the latter appearing for the first time towards the end of the Early Holocene (Supplementary Figure SI8). Unidentified charred plant fragments are also present, totalling 46 remains.

3.1.1.2 Middle Holocene: layer V (8.35–7.5 cal ka BP)

During this period, 6,242 carbonised seed remains were recorded, representing 12 identified taxa. Palms dominate the assemblage with 6,203 specimens, particularly *Syagrus orinocensis*, *Attalea maripa*, *A. insignis*, and *Astrocaryum chambira*, which are the most frequent species of the Middle Holocene. Other palms with low or absent representation include *Bactris* sp., *Euterpe* sp., *Mauritia* sp, and *Oenocarpus* sp. The Cocoseae tribe (Palmae) remains to constitute the largest palm group for this period, totalling 4,263 specimens. Arboreal taxa with low representation include *Brosimum lactescens* (Moraceae), *B. excelsa* (Lecythidaceae), and *Pera arborea* (Peraceae).

Unidentified seeds are rare, with only 20 remains recorded.

3.1.1.3 Late Holocene: layers VI–VII–VIII (1.9–1.4 cal ka BP; 1,550–1,480 cal AD; 1,560 cal AD)

A total of 24 taxa were identified, corresponding to 23,866 carbonised botanical remains. In the surface Layer VIII, 14 dry seeds of *Croton* sp. (Euphorbiaceae) and *Guatteria* sp. (Annonaceae) were recovered, suggesting that these species belong to the modern vegetation associated with the CM1 rockshelter.

The Palmae family is the most prominent taxonomic group throughout the entire archaeological sequence at CM1, and particularly during this period. The most abundant palms—*Syagrus orinocensis*, *Attalea maripa*, *A. insignis*, and *Astrocaryum chambira*—are represented by 9,764 endocarps. Fragmented seeds of palms of *Bactris* sp., *Mauritia* sp., *Euterpe* sp., and *Oenocarpus* sp. occur in low quantities, totalling 308 carbonised remains. Notably, *Mauritia* sp. is represented by both seed fragments and carbonised epicarp scales from the fruit. The palm tribe Cocoseae is the most abundant group in this period, with 13,239 identified archaeological remains.

Only a limited number of arboreal seeds associated with the Late Holocene were recovered. Identified taxa include *Sacoglottis mattogrossensis* (Humiriaceae), represented by 19 carbonised seeds and first appearing in Layer VI; *Hymenaea courbaril* (Fabaceae), with 5 seeds from Layers VII–VIII; *Sapium* sp. (Euphorbiaceae), with 5 carbonised seeds recovered in Layer VIII and associated with recent occupations at CM1; and *Pera arborea* (Peraceae), with a total of 46 seeds, comprising 15 dry seeds from the upper portion of Layer VIII and 31 carbonised seeds from Layers VI–VII, where the highest concentrations occur. Additionally, 24 seeds of *B. lactescens* (Moraceae) were identified, representing the largest quantity of remains for this species, and 25 seeds of *Byrsonima* cf. *spicata* (Malpighiaceae), which appear for the first time in the lower portion of Layer VI. Three fragments of copal resin (*Protium* sp., Burseraceae) were also recovered, exclusively associated with Late Holocene human occupations.

These resin pieces are characterised by a sub-rounded shape, translucent amber colouration, a preserved aromatic scent, and a thin, easily detachable whitish coating.

Towards the end of the Late Holocene, five carbonised fragments of maize (*Zea mays,* Poaceae) kernels were recovered, together with 352 carbonised unidentified seed fragments (morphotypes). This period shows the highest number of such undetermined remains, with two major peaks occurring in Layers VI–VIII.

3.1.2 Cerro Montoya 1. Unit 3, Grid E2

The plant macroremains from Unit 3, Grid E2, provide a complementary record to those from Unit 2, enabling assessment of intra-site variability in plant use over time, as described below (Figures 4, 5).

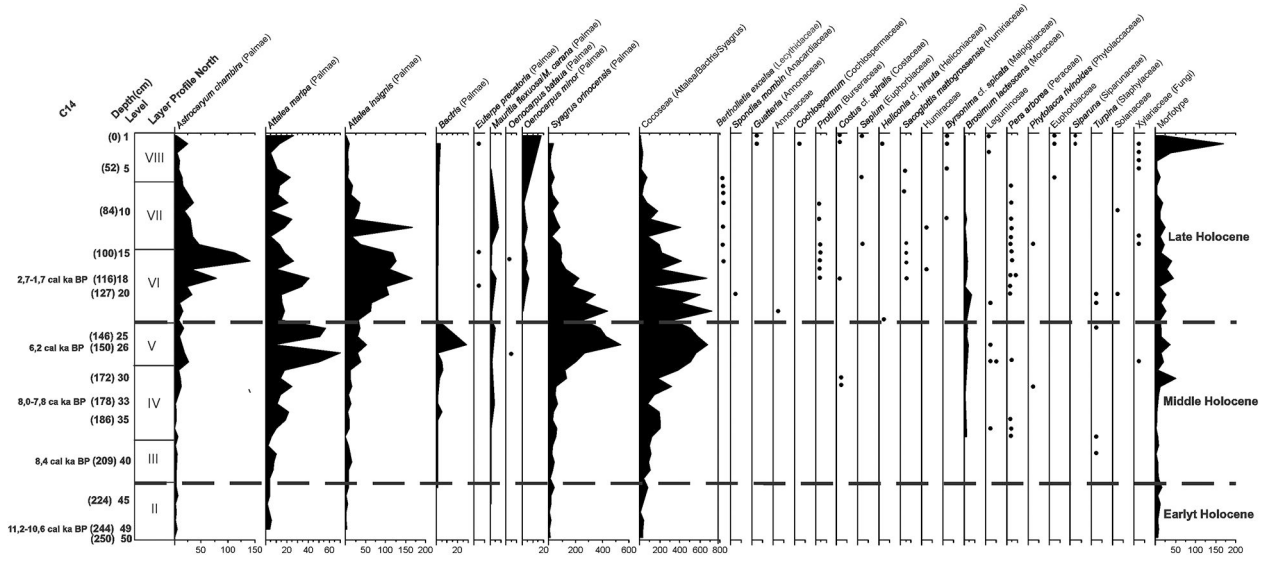


FIGURE 4
Stratigraphic diagram of seed counts from Cerro Montoya 1, Unit 3, Grid E2.

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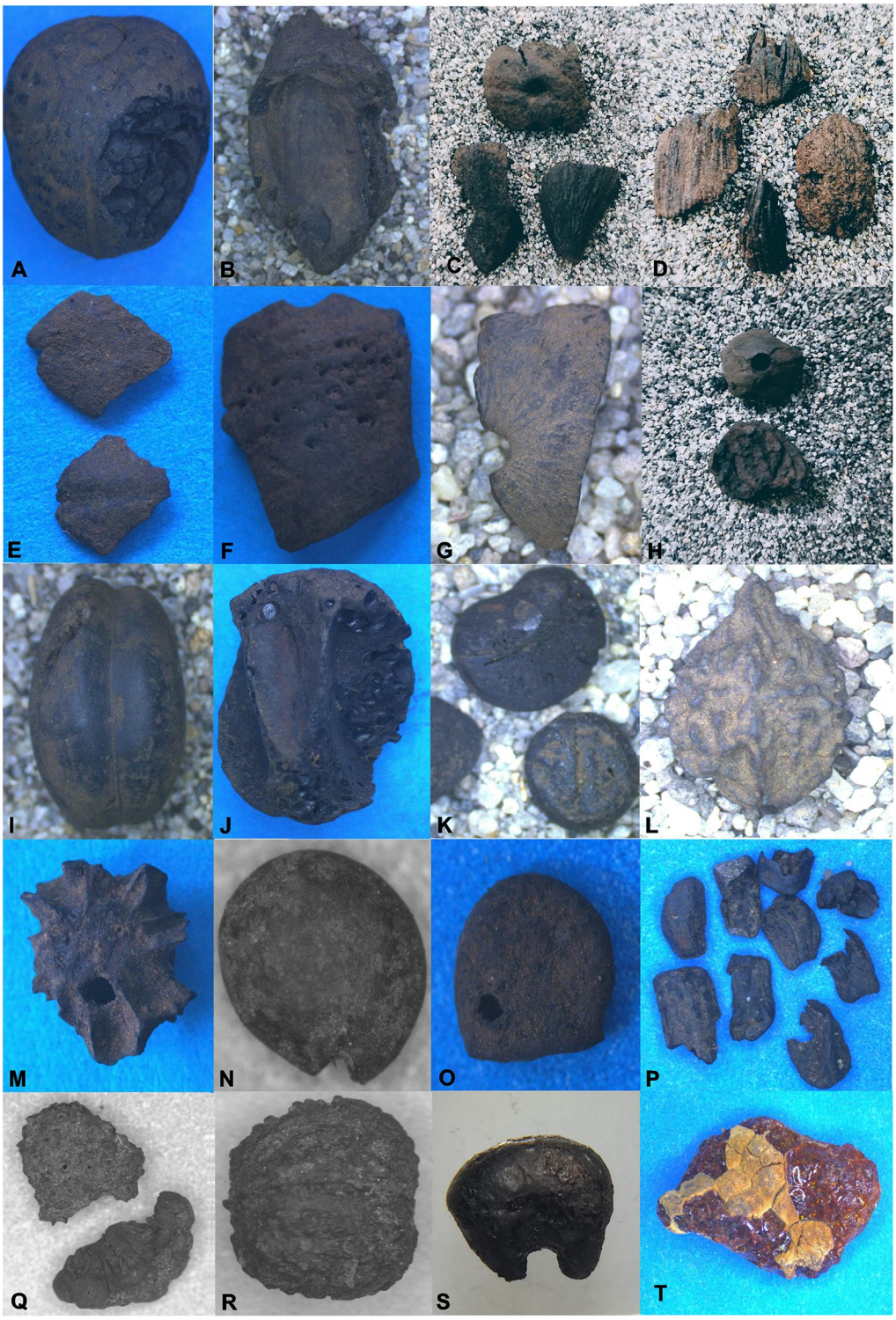


FIGURE 5
Carbonised archaeological macrobotanical remains from Cerro Montoya 1, Unit 3. **(A)** *Oenocarpus bataua* (Palmae); maximum length 18.5 mm (Layer VII). **(B)** *Attalea maripa* (Palmae); maximum length 20 mm (Layer III; 8.4 cal ka BP). **(C)** *Astrocaryum chambira* (Palmae), apices and germination pore; maximum length 10 mm (Layer IV; 8.0–7.8 cal ka BP). **(D)** *Attalea insignis* (Palmae); maximum length 15 mm (Layer V; 6.2 cal ka BP). **(E)** *Mauritia flexuosa* (Palmae), fruit scales (Layer VI; 2.7–1.7 cal ka BP). **(F)** *Syagrus orinocensis* (Palmae), endocarp fragment; maximum length 25 mm (Layer II; 11.2–10.6 cal ka BP). **(G)** *Bactris* sp. (Palmae), endocarp; maximum length 7.6 mm (Layer VI; 2.7–1.7 cal ka BP). **(H)** *Mauritia* sp. (Palmae); maximum length 25 mm (Layer IV; 8.0–7.8 cal ka BP). **(I)** *Hymenaea courbaril* (Fabaceae); maximum length 14.3 mm (Layer VII). **(J)** *Sacoglottis mattogrossensis* (Humiriaceae); maximum length 19 mm (Layer VI; 2.7–1.7 cal ka BP). **(K)** *Brosimum lactescens* (Moraceae); maximum length 8.2 mm (Layer VII). **(L)** *Pera arborea* (Peraceae); maximum length 10 mm (Layer VII). **(M)** *Byrsonima spicata* (Malpighiaceae); maximum length 7.1 mm. **(N)** *Phytolacca rivinoides* (Phytolaccaceae); maximum length 2 mm (Layer VI; 2.7–1.7 cal ka BP). **(O)** *Turpinia* sp. (Staphyleaceae); maximum length 3.2 mm (Layer VI; 2.7–1.7 cal ka BP). **(P)** *Costus* cf. *spiralis* (Costaceae) (Layer VI; 2.7–1.7 cal ka BP). **(Q)** *Xylariaceae* (Fungi) (Layer VI; 2.7–1.7 cal ka BP). **(R)** *Sapium* sp. (Euphorbiaceae); maximum length 6.6 mm (Layer VII). Cerro Montoya 1, Unit 2: **(S)** *Zea mays* (Poaceae); maximum length 7 mm (Layer VIII; 1,560 cal AD). **(T)** *Protium* sp. (Burseraceae), resin fragment; maximum length 15.7 mm (Layer VI; 1.9–1.4 cal ka BP).

3.1.2.1 Early Holocene: layer II (11.2–10.6 cal ka BP)

For the earliest occupational period, associated with the first inhabitants of CM1, 515 seed remains were recovered, comprising seven identified taxa, all palms. The most abundant species are *Syagrus orinocensis* (Palmae), represented by 113 endocarps, followed by *Astrocaryum chambira* (Palmae) with 23 remains, *Attalea insignis* (Palmae) with 21 remains, and *A. maripa* (Palmae) with 16 remains. Other palms with low representation include *Bactris* sp. and *Mauritia flexuosa*/*M. carana*. A total of 292 carbonised remains belonging to the Cocoseae tribe, including *Attalea* sp., *Astrocaryum* sp., and *Syagrus* sp., were identified.

3.1.2.2 Middle Holocene: layers III–IV–V (8.4 cal ka BP; 8.0–7.8 cal ka BP; 6.2 cal ka BP)

A total of 10,096 archaeological seeds were recovered for this period, with 13 identified taxa and a strong predominance of palms, represented by 8,654 remains. The Cocoseae tribe (*Attalea* sp., *Astrocaryum* sp., *Bactris* sp., *Syagrus* sp.) is the most prominent group, with 5,102 remains, followed by *Syagrus orinocensis* (2,535), *Attalea maripa* (478), *A. insignis* (320), and *Astrocaryum chambira* (147). Other palms, such as *Bactris* sp., *Mauritia flexuosa*/*M. carana*, and *Oenocarpus bataua*, are present in low frequencies.

Layer V contains the highest concentration of palms in the entire archaeological sequence, including Cocoseae (2,826), *Syagrus orinocensis* (1,817), *Attalea maripa* (307), and *Bactris* sp.(34).

Arboreal taxa with low representation during the Middle Holocene include *Costus* cf. *spiralis* (Costaceae), Fabaceae, and *Turpinia* (Staphyleaceae). The herbaceous species *Phytolacca rivinoides* (Phytolaccaceae), represented by a single carbonised seed, appears for the first time in Layer IV, while fungal remains belonging to Xylariaceae and associated with decaying wood appear for the first time in Layer V (Figure 5).

3.1.2.3 Late Holocene: layers VI–VII–VIII (2.7–1.7 cal ka BP)

The Late Holocene exhibits the highest concentration of botanical remains, totalling 17,603 specimens, and the greatest taxonomic diversity, with 28 identified archaeological taxa and two modern taxa.

As in earlier periods, palms dominate the assemblage, including *Syagrus orinocensis* with 2,537 remains, *Attalea insignis* with 1,156, *Astrocaryum chambira* with 740, *Attalea maripa* with 383, and the Cocoseae tribe, which is the most abundant group with 4,975 remains. Palms represented by fewer than 50 specimens include *Oenocarpus minor*, *Mauritia flexuosa*/*M. carana*, *Bactris* sp., *Oenocarpus bataua*, and *Euterpe precatoria*. The lower portions of Layers VI and VII show the highest concentrations of *Astrocaryum chambira*, *Attalea insignis*, and Cocoseae.

Several arboreal taxa appear at CM1 for the first time during this period and in low abundance, including *Bertholletia excelsa* (Lecythidaceae), *Spondias mombin* (Anacardiaceae), Annonaceae, *Protium* sp. (Burseraceae), *Costus* cf. *spiralis* (Costaceae), *Sapium* sp. (Euphorbiaceae), *Sacoglottis mattogrossensis* (Humiraceae), *Byrsonima* cf. *spicata* (Malpighiaceae), *Brosimum lactescens* (Moraceae), Fabaceae, *P. arborea* (Peraceae), *Siparuna* sp. (Siparunaceae), and *Turpinia* (Staphyleaceae). Two herbaceous species, *Heliconia* cf. *hirsuta* (Heliconiaceae) and *Phytolacca rivinoides* (Phytolaccaceae), are also present at low frequencies, together with one fungal taxon belonging to Xylariaceae.

3.2 Charcoal

The charcoal analysis presented here is based exclusively on the quantification and stratigraphic distribution of undifferentiated wood charcoal fragments recovered from the excavation units (Figure 6). No taxonomic identification of the charcoal was undertaken at this stage. Consequently, the results are used primarily as proxies for fire use, the intensity of domestic activities, and site-occupation dynamics over time. Detailed anthracological analyses aimed at identifying wood taxa, fuel selection, and vegetation management practices are planned for future work and will be necessary to refine interpretations of human–environment interactions at CM1.

Charcoal was recovered throughout the entire stratigraphic sequence at CM1, indicating persistent fire use and domestic activity across all phases of occupation. Fragment size decreases with depth in both units, suggesting strong taphonomic effects in the earliest levels. Charcoal concentrations are consistently higher in Unit 2, located within the rockshelter, indicating greater intensity of domestic activities in this protected space. Diachronically, the lowest charcoal values occur during the Late Pleistocene–Early Holocene occupations, followed by a general increase through the Holocene, with marked peaks in the Late Holocene, coinciding with high densities of archaeological seeds. Unit 3 shows similar trends but with lower overall charcoal volumes, reflecting differential use of internal and external activity areas (Figure 6).

3.3 Phytoliths

One proxy that extends the range of plant taxa identified at the CM1 archaeological site is phytolith analysis, as these exceptionally well-preserved microfossils occur at high densities and broaden our understanding of human–plant relationships throughout the sequence of human occupation at the site. In the following section, the phytolith assemblage from Unit 3, Grid E2 is described chronologically to highlight patterns of predominance, stability, and change in plant exploitation over time. A total of 18 taxa were identified, with the assemblage dominated by the families Palmae and Poaceae (Figure 7 and Supplementary Figure SI12), followed by representatives of the order Zingiberales, whereas arboreal taxa are generally underrepresented.

3.3.1 Cerro Montoya 1— unit 3, grid E2

The CM1, Unit 3, Grid E2 phytolith diagram depicts taxa associated with human groups over the period from 11.2 cal ka BP to the Late Holocene (Figure 7).

3.3.1.1 Early Holocene: layer II (11.2–10.6 cal ka BP)

During this period, one of the highest palm phytolith densities recorded at the site is observed. The most prominent group is the Cocoseae tribe (*Attalea*/*Syagrus*), accounting for 22%–65% of the

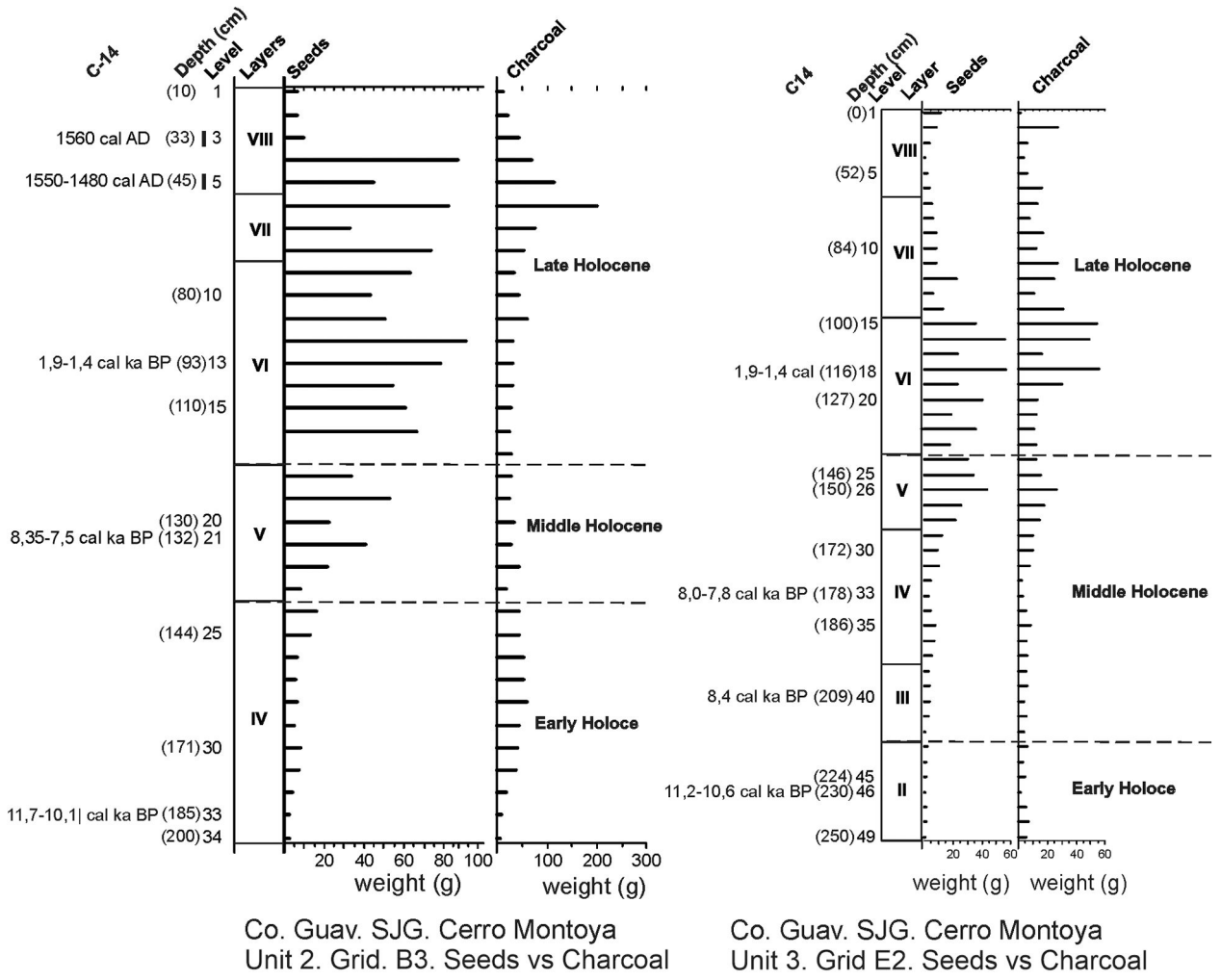


FIGURE 6
Charcoal/weight distribution from Cerro Montoya 1. Left: Unit 2, Grid B3. Right: Unit 3, Grid E2.

assemblage, followed by *Astrocaryum*/*Bactris* (5%–30%) and *Euterpe*/*Oenocarpus* (3%–30%). *Geonoma* sp. and *Mauritia* sp. occur at consistently low frequencies (≤5%) (Figure 8). Moderate representation is recorded for Poaceae (15%–30%), the order Zingiberales (Costaceae/Marantaceae/Strelitziaceae) (5%–15%), and *Heliconia* sp. (5%–10%). Phytoliths from Cyperaceae, as well as Morphotype 7 and Morphotype 8, are rare during this phase.

3.3.1.2 Middle Holocene: layers III–IV–V (8.4 cal ka BP; 8.0–7.8 cal ka BP; 6.2 cal ka BP)

This period, spanning slightly more than 2,000 years, shows a pattern broadly similar to that observed in the Early Holocene. Palms continue to dominate the assemblage, particularly the Cocoseae tribe (30%–50%). Phytoliths of *Astrocaryum*/*Bactris* (5%–30%) decrease markedly in the upper portion of Layer IV and in Layer V, while *Euterpe*/*Oenocarpus* (5%–30%) show low representation in Layer III. *Geonoma* sp. and *Mauritia* sp. remain present at frequencies below 5%. Herbaceous taxa are moderately represented, including Zingiberales (10%–15%), Poaceae (5%–25%), and Heliconiaceae (≤10%). Very low frequencies are recorded for Cyperaceae, arboreal Annonaceae, and Morphotypes 5, 7, and 8.

3.3.1.3 Late Holocene: layers VI–VII–VIII (2.7–1.7 cal ka BP)

Palm taxa continue to show high representation, with the Cocoseae tribe reaching its highest values across the entire occupational sequence (35%–60%). In contrast, *Astrocaryum*/*Bactris* exhibits its lowest frequencies of the sequence (3%–10%). *Euterpe*/*Oenocarpus* (Palmae) (5%–30%) maintains values comparable to those of the Middle Holocene, while *Geonoma* sp. and *Mauritia flexuosa*/*M. carana* (Palmae) remain weakly represented.

Herb-associated taxa display notable changes during this period. Zingiberales (Costaceae/Marantaceae/Strelitziaceae) reach their highest frequencies at the site (5%–35%), whereas grasses (Poaceae) maintain generally low values (8%–20%), and only a small number of Cyperaceae phytoliths are recorded. In Layer VI, phytolith structures attributed to *Licania* sp. appear for the first time (Supplementary Figure SI13); these derive from arboreal taxa whose bark is used as ceramic temper by some groups in the Amazon basin.

Phytoliths of manioc (*Manihot esculenta*, Euphorbiaceae), characterised by their distinctive heart-shaped morphology, were identified in association with the Late Holocene (2.7–1.7 cal ka BP) (Supplementary Figures SI9, SI10). This interval corresponds to the presence of ceramics at the CM1 site after the late Middle Holocene hiatus. The new occupants incorporated ceramic

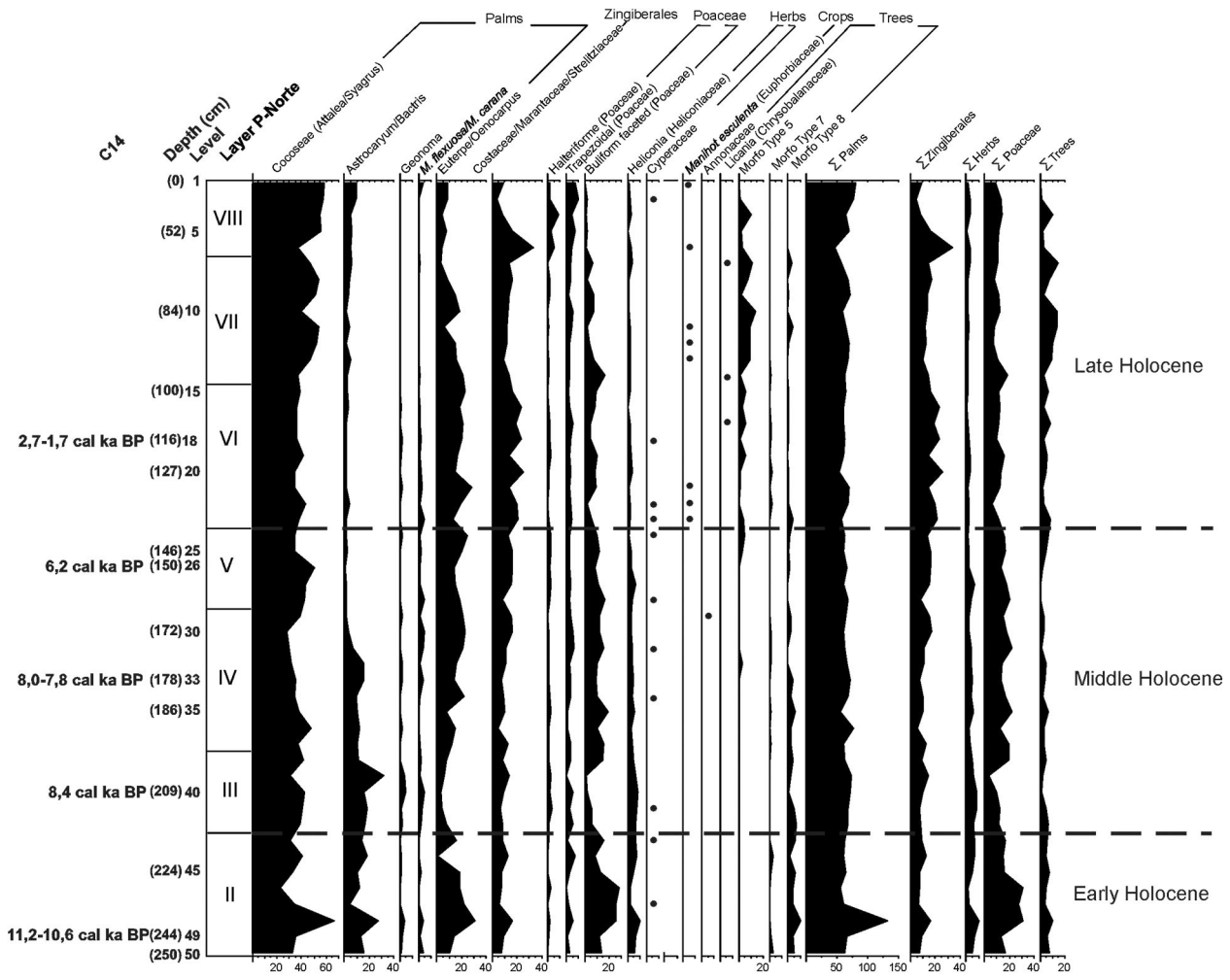


FIGURE 7
Percentage stratigraphic diagram of phytolith taxa from Cerro Montoya 1, Unit 3, Grid E2.

technology and plant cultivation, indicating the integration of cultivated root crops into plant-use practices during this phase, alongside the continued exploitation of wild plant resources.

3.4 Starch grains

A total of 311 starch grains were identified in samples dated from the Early Holocene (~11.3 cal ka BP), evidencing the importance of carbohydrate-rich plants and the broad spectrum of taxa exploited (Figure 9, Table 2). Identifications were based on reference collections and on taxa previously recorded through macrobotanical and phytolith analyses, as these represent plants known to have been used at the site.

Certainty levels vary, since several morphotypes are compatible with multiple taxa, leaving taxonomic attribution open to more than one species. Another limitation is that several wild rhizomatous plants reported from the study area were not collected for reference material, and therefore as their morphology is unknown, they cannot be excluded. Approximately 47% of the starches were identified to some degree, indicating that many grains derive from plants of potential use that remain unidentified but reinforce the notion of a broad-spectrum plant economy at SLL.

3.4.1 Fabaceae (legumes)

Thirty-four starches are compatible with Fabaceae, characterised by simple, oval, medium-to-large grains (15–37 μm), often showing lamellae, a central longitudinal fissure, and a symmetrical malt cross. Locally, *Inga* sp. (guamo negro), whose seeds enclosed in large pods are edible, produces starches consistent with those found at SLL (Piperno and Dillehay, 2008). *Inga* sp. starches are medium-sized (10–20 μm) with a central fissure and were recovered in sample CM2 212897. Another compatible species in the area is *Bixa orellana* (Bixaceae) (achiote), whose seeds yield medium-to-large oval starches lacking lamellae. Two samples from ceramic occupations (CM2_210412 and CM2_210258) contained assemblages compatible with *Hymenaea cf. courbaril* (Fabaceae) (algarrobo), a leguminous tree (Table 2). These starches are small (<10 μm), bell-shaped or polyhedral, with an open hilum, a central depression, and a symmetric malt cross. The species *Hymenaea oblongifolia*, a member of the same genus, is also reported in the region.

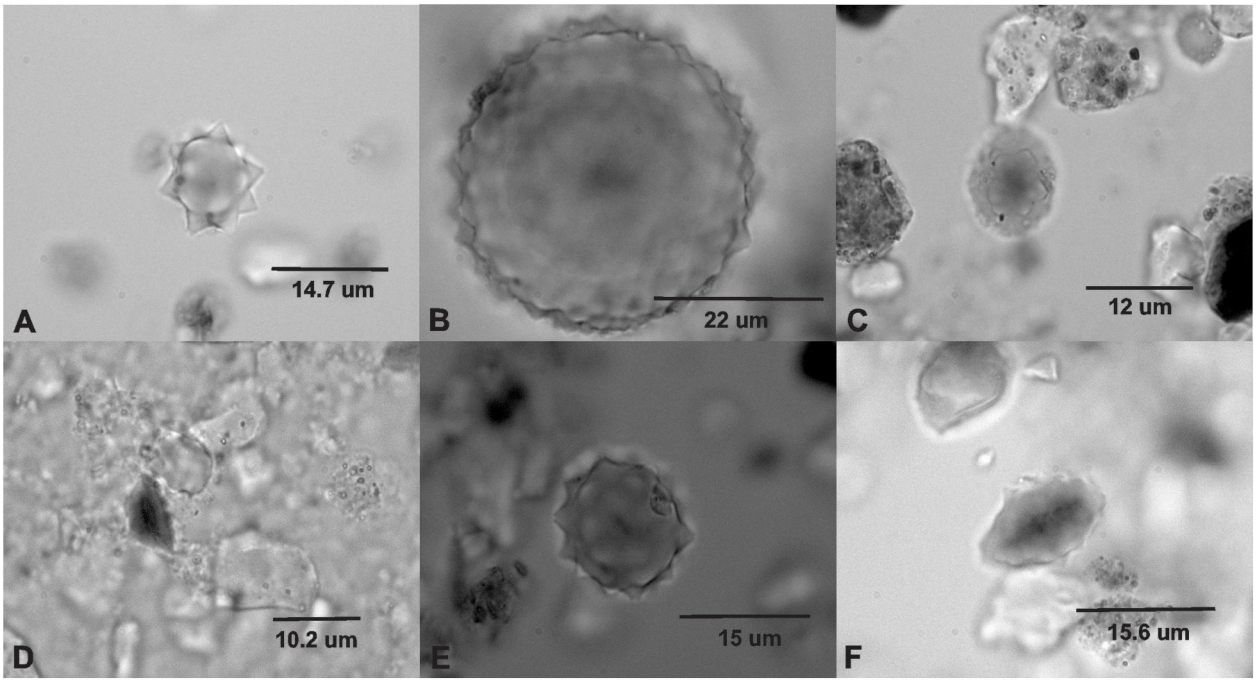


FIGURE 8
Archaeological palm phytoliths from CM1.**(A)** *Mauritia flexuosa/Mauritia carana* (Palmae), symmetrical globular echinate phytolith (Unit 3, Layer VIII). **(B)** *Euterpe/Oenocarpus* (Palmae), globular echinate phytoliths with short sharp projections (Unit 3, Layer VII). **(C)** *Astrocaryum/Bactris/Iriartella/Socratea*, apical view: conical phytoliths with ruminate margin and central projections (Unit 3, Layer VII). **(D)** *Astrocaryum/Bactris/Iriartella*, frontal view: conical phytoliths (Unit 3, Layer II: 11,2-10,6 cal ka BP). **(E)** *Attalea/Syagrus*, globular echinate phytoliths (100X) (Unit 3, Layer VI: 2,7-1,7 cal ka BP). **(F)** *Geonoma* sp., conical phytoliths with basal projections (Unit 3, Layer III: 7,97-7,79 cal ka BP).

3.4.2 Rhizomatous plants

Sixty-two starches (41 from preceramic and 21 from ceramic occupations; ~20% of total) were associated with rhizome-bearing taxa (Table 2). *Costus cf. spiralis* (Costaceae) was identified by medium-to-large (19–38 μm), pyriform or truncated-pyramidal grains with an eccentric hilum, visible lamellae, and an asymmetric malt cross; eight starches of this taxon were identified, three from a Middle Holocene sample. Six starches were identified as *Heliconia acuminata* (Heliconiaceae), with diagnostic medium (19–38 μm), elongated pyriform grains showing marked lamellae, eccentric hilum, and asymmetric malt cross; five derive from a Middle Holocene sample and one from an undated context. *Heliconia acuminata* is a wild herb native to the area that produces soft, carbohydrate-rich rhizomes that are potentially edible. Seventeen additional starches were assigned to *Costus cf. spiralis/Heliconia cf. psittacorum*, as both share this morphotype: large (>20 μm), elongated grains with straight distal ends. Two came from Early Holocene contexts, thirteen from Middle Holocene contexts, and two from Late Holocene contexts (Table 2). The presence of these starches on handstones suggests rhizome processing. *Costus spiralis* is a native herb with soft, edible rhizomes, and *Heliconia* spp. are abundant in the region and have known ethnobotanical uses: the Nukak use the leaves of *H. psittacorum*, *H. stricta*, and *H. hirsuta* to make baskets (Cárdenas and Politis, 2000), while *H. hirsuta* rhizomes, which resemble manioc, are edible and consumed among the Huitoto of Perú (Martínez and Galeano, 1994). This species also appears in the phytolith record, suggesting that the *Heliconia* spp. starches could correspond to *H. hirsuta*, for which starch reference samples are not yet available.

3.4.3 Underground storage organs (USO)

Twelve large grains correspond to USO-type morphologies, a key carbohydrate source (Table 2). Five resemble *Dioscorea* spp. (Dioscoreaceae) (yams); species recorded in the area include *D. atrescens* and *D. guianensis*, for which ethnobotanical use is unrecorded but cannot be excluded. The genus *Dioscorea* is rich in nutraceutical compounds with antioxidant, antibacterial, and antiallergenic properties (Salehi et al., 2019). Similar morphotypes were also observed in *Philodendron* sp. (Araceae) and *Calathea* sp. (Marantaceae); local species *P. acutatum* and *C. cyclophora* both have rhizomes, and *P. victoriae* is used medicinally by the Nukak (Cárdenas and Politis, 2000). These morphotypes occur in both preceramic (Middle Holocene) and ceramic contexts (Table 2). Two large grains from sample 210414 (ceramic context) with marked lamellae and distinctive appendices match *Phenakospermum guyannense* (Strelitziaceae), suggesting this native plant despite some morphological differences.

3.4.3 Marantaceae

Thirteen starches are consistent with *Maranta arundinacea* (arrowroot) (Table 2). These are medium-sized (11–23 μm), oval, with eccentric fissured hilum and faint lamellae—diagnostic for this taxon. Although not locally recorded, *Maranta* spp. are Neotropical rhizomatous herbs distributed across the Colombian

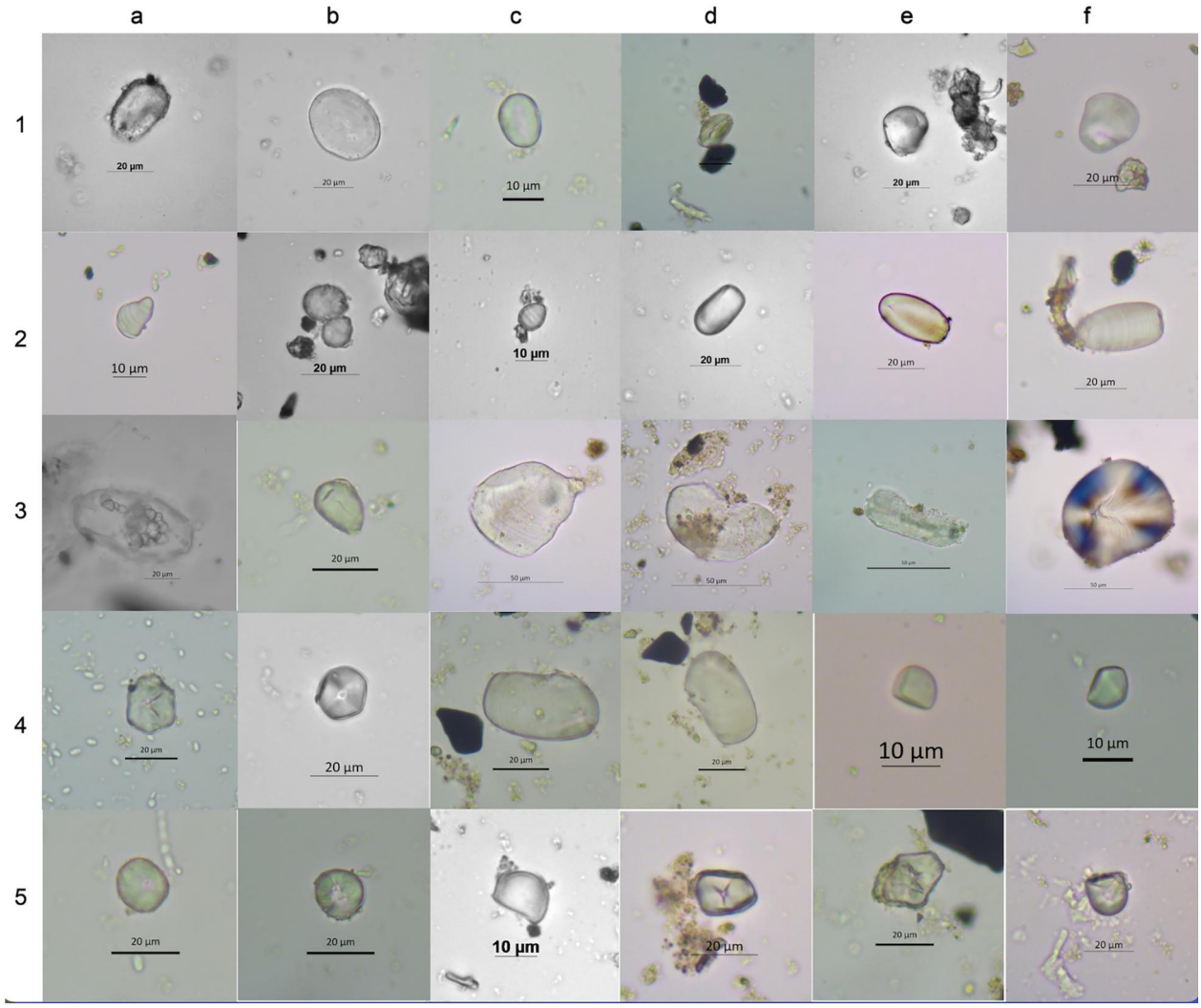


FIGURE 9
Selected archaeological starch grains from Cerro Montoya 1. **(1a)** Fabaceae (L211368) (~10.2 cal ka BP); **(1b)** Fabaceae (L210536) (~1.8 cal ka BP); **(1c)** *Inga* spp. (Fabaceae) (L213415) (~7.9 cal ka BP); **(1d)** Fabaceae (L212018) (~8.2 cal ka BP); **(1e,f)** *Costus cf. spiralis* (Costaceae) (L211368) (~10.2 cal ka BP). **(2a)** *Heliconia cf. acuminata* (Heliconiaceae) (L210536) (~1.8 cal ka BP); **(2b)** *Heliconia cf. acuminata* (L214068) (no date); **(2c)** *Heliconia cf. acuminata* (L211828) (~11.2 cal ka BP); **(2d)** *Heliconia cf. psittacorum* (L211368) (~10.2 cal ka BP); **(2e,f)** *Heliconia cf. psittacorum* (L213415) (~7.9 cal ka BP). **(2b)** *Hymenaea cf. courbaril* (Fabaceae) (L210258) (undated); **(3b)** *Maranta cf. arundinacea* (Marantaceae) (L213415) (~7.9 cal ka BP); **(3c)** Underground storage organ (USO) (L210414) (~1.8 cal ka BP); **(3d)** USO (L210018) (~8.2 cal ka BP); **(3e,f)** USO (L213415) (~7.9 cal ka BP). **(4a)** Polyhedral morphotype (L211368) (~10.2 cal ka BP); **(4b)** Polyhedral morphotype (L210536) (≈1.8 cal ka BP); **(4c,d)** *Dioscorea* spp. (Dioscoreaceae) (L212876) (~2.5 cal ka BP); **(4e)** Small bell-shaped morphotype (cf. *Xanthosoma* sp., *Monotagma* sp., *Attalea* sp., *Bactris* sp.) (L210536) (~1.8 cal ka BP); **(4f)** Small bell-shaped morphotype (L212876) (~2.5 cal ka BP). **(5a,b)** *Theobroma cf. obovatum* (Malvaceae) (L213415) (~7.9 cal ka BP); **(5c)** *Monotagma cf. laxum* (Marantaceae) (L211368) (~10.2 cal ka BP); **(5d)** *Zea mays* (Poaceae) (L210414) (~1.8 cal ka BP); **(5e)** *Zea mays* (L212876) (~2.5 cal ka BP); **(5f)** *Manihot esculenta* (Euphorbiaceae) (L210414) (~1.8 cal ka BP).

Amazon (Brito et al., 2021). In Araracuara, several *Maranta* species are documented, including edible taxa such as *M. amplifolia*, cultivated by the Uitoto, and *M. ruiziana*, grown for its rhizomes (Suárez and Galeano, 1996). Thus, the use of a *Maranta* species other than *M. arundinacea* cannot be ruled out. Eight starches derive from Early–Middle Holocene preceramic contexts and five from a context dated to 1600 CE (Table 2).

3.4.4 Grasses and palms

A common morphotype ($n = 46$) comprises medium (10–20 µm), faceted, polyhedral grains with a central fissured hilum, characteristic of several grasses and cultivated palms (Table 2). Among regional plants, starches of *Euterpe cf. precatoria* (Palmae) match this morphology; this palm was also identified through macrobotanical remains and phytoliths. However, other useful plants not represented in the reference collection cannot be excluded. These morphotypes occur in Early and Middle Holocene samples (Table 2).

3.4.5 Small morphotypes

Fourteen small bell-shaped and six small polyhedral grains (6–10 µm) correspond to several taxa, including *Xanthosoma* sp. (Araceae), *Monotagma laxum* (Marantaceae), *Theobroma*

obovatum (Malvaceae) (wild cacao), and palms (*Attalea* sp., *Acrocomia* sp., *Bactris* sp.) and *Zamia* spp. (Table 2).

3.4.6 Maize (*Zea mays,* Poaceae)

Thirty-seven medium (12–20 μm) polyhedral grains, angular and multifaceted, with fissured hilum—one of the most diagnostic traits —are consistent with maize (Table 2). Half of these were recovered from the base of a vessel containing charred macrobotanical remains, dated to 420 cal BP (Aceituno et al., 2024a).

3.4.7 Cacao (*Theobroma cf. obovatum*, Malvaceae)

Eleven medium (10–16 μm) oval grains with central open hilum, sometimes fissured, match wild cacao starches (Table 2). This local tree species has an edible mesocarp. Except for two grains from Middle Holocene levels, all derive from lithic assemblages associated with ceramic occupations.

3.4.8 Other taxa

Three medium, bifaceted, bell-shaped grains are compatible with *Monotagma cf. laxum* (Marantaceae), a native marantaceous herb with a small rhizome, though no ethnobotanical use is known in the Colombian Amazon (Suárez and Galeano, 1996). These derive from an Early Holocene sample (Table 2). One medium bell-shaped grain from a ceramic occupation is diagnostic of manioc (*Manihot esculenta,* Euphorbiaceae) (Table 2).

In summary, starch evidence from CM1 reveals temporal variability in plant use spanning from the Late Pleistocene–Early Holocene to the Late Holocene (Table 2). The earliest occupations (Late Pleistocene–Early Holocene) already show the exploitation of carbohydrate-rich rhizomatous taxa, notably *Costus cf. spiralis* (Costaceae) and *Heliconia cf. psittacorum* (Heliconiaceae), together with morphotypes compatible with Fabaceae and USO-type starches, indicating a broad-spectrum subsistence strategy.

During the Middle Holocene, the assemblage diversified to include *Heliconia acuminata* (Heliconiaceae), *Maranta arundinacea* (Marantaceae), *Philodendron* sp. (Araceae) and *Calathea* spp. (Marantaceae), alongside continued use of Costus- and Dioscorea-like plants, suggesting an intensified reliance on edible rhizomes and underground storage organs. The Late Holocene record is characterised by the introduction or increased visibility of cultivated and managed species, including maize (*Zea mays*), manioc (*Manihot esculenta*), and wild cacao (*Theobroma cf. obovatum*), alongside persistent use of palm taxa (*Euterpe cf. precatoria*) and other forest resources. This diachronic pattern reflects a long-term continuity in the exploitation of carbohydrate-rich plants, gradually complemented by the incorporation of domesticated and managed forest products into the subsistence systems of La Lindosa's inhabitants.

3.5 Seasonality of resources

To explore the seasonal availability of these resources, we present a first approximation of the annual fruiting cycles of eleven wild food species, including palms, fruit trees, and other food-bearing taxa (Figure 10). Based on herbarium data and field observations (Supplementary Table SI4), fruiting peaks for palms and food fruit trees in the Colombian Amazon occur primarily between January and June, spanning both the dry and early rainy seasons, with a decline during the period of lower rainfall from July to December.

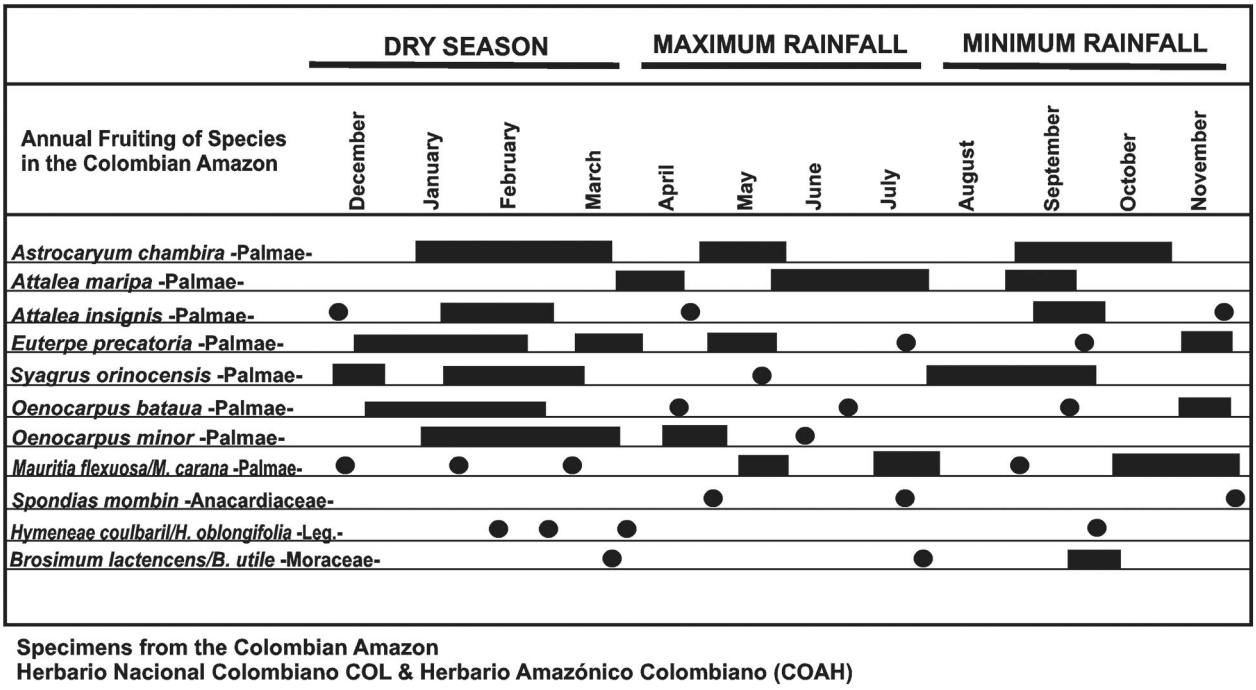


FIGURE 10
Fruiting cycles of plant species identified at Cerro Montoya 1.

Notably, the four palm species most prominently represented in the archaeobotanical record (*Syagrus orinocensis*, *Attalea insignis*, *A. maripa*, and *Astrocaryum chambira*) bear fruit over much of the annual cycle. This extended and predictable availability likely contributed to their preferential selection by past human groups. More generally, the graph shows that palm resources were available for most of the year, with only limited seasonal gaps, and that the taxa most abundant in the archaeobotanical record correspond to those with the longest and most continuous fruiting periods.

In contrast, other palms with high food potential (*Mauritia flexuosa*, *Oenocarpus bataua*, *O. minor*, and *Euterpe precatoria*), which also fruit over extended periods, are poorly represented at CM1. A plausible explanation for this pattern is a low local density of these species in the SLL, which reduces availability despite their high nutritional value. Overall, these patterns suggest that availability and predictability were key factors shaping palm selection, reinforcing the interpretation that palms constituted reliable, year-round resources within the subsistence strategies of the inhabitants of CM1.

4 Discussion and conclusions

As demonstrated by our results, the multiproxy approach applied at CM1—integrating plant macroremains (fruits, seeds, charcoal, and resins) with phytolith and starch-grain analyses—overcomes the interpretative limitations of any single method and enables a robust reconstruction of plant-use practices among the inhabitants of the Serranía de La Lindosa. Of particular significance is the incorporation of plant microfossil analyses (phytoliths and starch grains), which are essential for identifying rhizomes and other vegetatively reproducing tuberous plants—taxa that are extremely difficult to detect through the exclusive analysis of macrobotanical remains (Fuller et al., 2023). Together, these complementary datasets document a wide spectrum of plant exploitation, including dietary, technological, medicinal, and other utilitarian applications, from the initial Late Pleistocene occupation through to the colonial period.

Human colonisation of the SLL during the Late Pleistocene to Early Holocene (12.6–10.0 cal ka BP) was associated with seven palm species and one tree species: *Syagrus orinocensis*, *Attalea insignis*, *A. maripa*, *Astrocaryum chambira*, *Bactris* sp., *Mauritia flexuosa*, *M. carana*, *Oenocarpus minor*, and *Brosimum lactescens* (Moraceae). Charred root parenchyma, phytoliths, and starch grains demonstrate that tubers also played a key role in early Amazonian diets. These analyses have identified several carbohydrate-rich taxa, including *Costus* cf. *spiralis* (Costaceae), *Heliconia acuminata* (Heliconiaceae), *Maranta arundinacea* (Marantaceae), and *Dioscorea* spp. (Dioscoreaceae), as well as other unidentified rhizomes (USO), all of which produce edible rhizomes or underground storage organs. Their consistent presence across preceramic and ceramic contexts highlights the long-term importance of root and tuber exploitation as a stable foundation of early subsistence in tropical forest environments.

The dominance of palms in the plant assemblage is unsurprising, as many Amazonian hunter-gatherer groups—such as the Nukak, Awá, and Hotï—have historically relied heavily on palms. These plants are characterised by high fruit productivity, protein- and oil-rich seeds, and extended fruiting periods that provide resources throughout the year. Ethnographic observations show that hunter-gatherers in South American tropical forests organise daily foraging expeditions specifically aimed at fruit collection, with resources either consumed immediately or transported back to camp in large quantities, often carried in containers or as entire bunches (Politis and Loperfido, 2025). This is consistent with our argument in Robinson et al. (2021) that palms were instrumental in enabling human dispersal into new tropical environments, providing visible, dependable, and low-risk resources during early colonisation.

Ethnobotanical studies among the Nukak, who inhabit the SLL region, further demonstrate the breadth of plant exploitation strategies (Cárdenas and Politis, 2000). In total, 113 plant species belonging to 44 families, including herbs, palms, trees, shrubs, and vines, have been recorded for diverse uses, of which 77 species serve as food sources. The most represented family is Palmae, with 15 species used not only for subsistence but also for a wide range of technological purposes, including the manufacture of carrying bags (cargueros), the use of trunk fibres for dart production, leaves for house construction, and trunks as substrates for larval rearing. Palms thus account for both the highest number of collected species and the greatest biomass yield, supporting their central role in daily life

Politis (2009). also shows that the Nukak actively shape their territories by increasing the abundance and visibility of useful plant species. Such practices likely contributed to the dominance of taxa highly valued by Indigenous peoples, including palms and other economically important species such as *Brosimum* sp (Cárdenas and Politis, 2000; Politis, 2009). As previously argued (Iriarte, 2024; Morcote-Ríos et al., 2014; Politis, 2009; Politis and Loperfido, 2025), the nutritional and technological advantages of palms, combined with their ubiquity, high density, and productivity, made them first-rank resources for the earliest hunter-gatherer populations occupying the lowlands of South America. As we have seen in section 3.5, fruiting period data indicate a year-round availability of palm fruits, resulting from staggered and partially overlapping fruiting periods among different palm taxa, with the dry season appearing to concentrate a higher number of fruiting taxa.

It is important to note that there is no direct archaeological or palaeoecological evidence allowing us to determine whether palms were deliberately managed in the past, or to identify the specific forms such management may have taken. At present, there are no data that allow us to specify the mechanisms behind this dispersal, whether through fire, forest clearance, or other management practices. Nevertheless, the archaeobotanical record suggests selective palm management, likely involving the dispersal of these fruit-bearing species and the formation of small palm "grooves". It is plausible that sustained and intensive harvesting practices contributed to the expansion of palm populations, either through unintentional dispersal mechanisms, such as those documented among Nukak groups today (Politis, 2009), or through deliberate actions, such as the creation of small clearings that favoured palm recruitment and growth. Even in the absence of direct evidence for these practices, the long-term, recurrent use of palms suggests that human groups may have deliberately encouraged their spread.

The establishment of new palm stands through seed dispersal may represent a pathway towards domestication

(Spengler, 2020; Spengler et al., 2025), since it can increase gene flow among wild populations, promote genetic and phenotypic variation, and foster mutualistic relationships, in this case between palms and humans (Spengler, 2020). At the same time, the encouragement, protection, dispersal, and care of useful non-domesticated species can also be understood as forms of cultivation, even in the absence of full domestication (e.g., Scheel-Ybert et al., 2022; Scheel-Ybert and Boyadjian, 2020). Such practices need not be framed within a rigid opposition between hunter-gathering and agriculture. Rather, the protection, dispersal, tending, and care of useful non-domesticated species may be seen as forms of cultivation that produce anthropogenic landscapes without necessarily implying domestication or field agriculture (Scheel-Ybert and Boyadjian, 2020; Scheel-Ybert et al., 2022). Moreover, the high taxonomic diversity of palms documented in the archaeobotanical record is consistent with the presence of mature, structurally intact forests (Balslev et al., 2017). In this sense, forest management by past groups in the SLL is likely to have been subtle and cumulative, operating over millennial timescales. Small palm groves may also have functioned as focal points for fauna, attracting species such as tapirs, deer, and peccaries, and may have facilitated hunting opportunities in a manner analogous to animal exploitation around cultivated fields (SLL, 1976). More broadly, the La Lindosa evidence is best interpreted as part of a long-term continuum of forest management and cultivation practices, comparable to other South American cases in which forests and gardens were historically entangled rather than sharply separated (Scheel-Ybert et al., 2016).

By contrast, for rhizomes, as noted above, and for the other identified plants, the available evidence does not allow us to confirm or reject a hypothesis of pre-domestication cultivation. However, the absence of domestication in the most representative taxa (e.g., Costus, Phenakospermum, and Heliconia), together with the lack of direct evidence for their cultivation, supports the interpretation that these plants were not cultivated. This does not exclude the possibility that they were favoured within a domesticated or anthropogenic forest, following the framework proposed by Charles R. Clement (Clement et al., 2024; Clement et al., 2021) and by William Balée and collaborators (Balée, 2010; Balée and Erickson, 2006; Balée and Gely, 1989), that is, ecosystems modified by human activity in ways that encourage particular plant and animal species.Noticeably, a substantial proportion of the hyperdominant species in Amazonian forests are palms that appear in the earliest archaeobotanical records and were intensively exploited by hunter-gatherers (see summary in Iriarte et al., 2020). Their sustained use and management over millennia may have played a key role in shaping their present-day hyperdominance (Iriarte et al., 2020; Levis et al., 2017; Ter Steege et al., 2013), a pattern further supported by the documented increase in palm exploitation through time in the archaeobotanical record.

Interestingly, examination of charred lumps of palm endocarp and mesocarp under the SEM at the nearby Limoncillos rockshelter in the SLL (Garay-Vázquez et al. in prep.) reveals collapsed cellular tissue with a homogenised, paste-like appearance. Such tissue deformation is consistent with mechanical processing, particularly pounding or grinding of palm kernels, rather than simple roasting or consumption of raw fruits.

These observations suggest that palms at Limoncillos were not merely eaten whole but were also processed into doughs, porridges, or fermented beverages such as chicha.

The collapse of endosperm tissues, therefore, provides indirect but compelling evidence for intensive culinary preparation and transformation of palm resources. The microstructural characteristics and experimental basis for this interpretation are discussed in detail in Garay-Vázquez et al. (in prep).

To what extent palm starch grains may reflect the exploitation of palm trunk starch—such as that documented among Tukanoan groups in the region using *Iriartea deltoidea* (Bernal et al., 2007), and more broadly across tropical and subtropical regions of the Americas with various palm species (see synthesis in Meggers, 2001) is difficult to demonstrate at present, but this possibility should not be dismissed.

When the archaeobotanical results from CM1 are considered alongside data from other archaeological sites across the Amazon basin, they offer a glimpse of what may be described as regionally distinct "cultural complexes of palm management" in Amazonia. In the Colombian Amazon, the Guiana Shield region —between the SLL and the Serranía de Chiribiquete—can be characterised by a *Syagrus–Astrocaryum–Attalea* complex, reflecting a preferential selection of *Syagrus orinocensis, Astrocaryum chambira*, and *Attalea insignis*. In contrast, evidence from the central Colombian Amazon (Middle Caquetá River), particularly at Peña Roja, points to an *Oenocarpus–Mauritia–Astrocaryum* complex, marked by the prominence of *Oenocarpus bataua*, *Mauritia flexuosa*, and *Astrocaryum chambira*. Further downstream, in the middle and lower Amazon River region of Brazil, available evidence suggests a third palm-management complex dominated by *Euterpe, Astrocaryum*, and *Attalea*, reflecting the long-term importance of these genera in Amazonian palm exploitation strategies. This aspect is further explored in Garay-Vázquez et al. (in prep).

SLL appears to represent a distinct case. In contrast to the early and regionally differentiated trajectories of plant domestication and incipient cultivation documented across north-western South America (Aceituno and Loaiza, 2018; Piperno 2011) and south-western Amazonia (Lombardo et al., 2020; Watling et al., 2018), the evidence from SLL points to a broadly generalised subsistence strategy during its initial occupation.

While neighbouring regions, such as the upper and middle Cauca, River, Calima River basin, and other areas of the Colombian Andes (e.g., Aceituno and Loaiza, 2018; Aceituno et al., 2022; de Schrimpff et al., 2019; Gnecco, 2003) and areas to the south in the Colombian Amazon exhibit early management and domestication process of tubers, rizhomes, tree fruits (i.e avocado) and squashes—often associated with forest clearance (landscape alteration or disturbance) and specialised cultivation tools, like axes or hoes (see summaries in Aceituno and Loaiza, 2018; Iriarte et al., 2020), SLL is characterised by diversified plant use without clear signals of early intensification or landscape modification or early predomestication cultivation. This situates SLL closer to other tropical forest contexts in which flexible, broad-spectrum economies, —with some degree of palm management—, were established from the outset, rather than among the early centres of sustained plant domestication identified elsewhere in the region.

Evidence for plant cultivation appears only in the Late Holocene, as indicated by the identification of manioc and maize starch grains. This interpretation is further supported by the recovery of maize phytoliths from the lower slope below Limoncillos rockshelter, within a black soil layer dated to the sixth century C.E. Comparable evidence has also been documented in other terra preta contexts near the tepuis where the rock shelters are located, including Cerro Montoya 1, with chronologies extending from the first millennium B.C.E. into the Common Era (Kosztura, 2020; Kosztura Nuñez and Morcote-Ríos, 2023). At Buena Vista, maize phytoliths have been identified in deposits dated between 775 cal B.C.E. and 973 C.E (Kosztura Nuñez and Morcote-Ríos, 2023).

The recovered starch assemblage provides evidence for the processing of rhizomes from wild plants, as well as fruits from leguminous trees, wild cacao, palms, and other plant resources. At present, we lack the archaeological and palaeoecological evidence needed to test or document whether rhizomatous plants were cultivated, and we recognise that such forms of plant manipulation are inherently difficult to identify in the archaeological record. In other tropical forest contexts, relatively simple forms of plant management have been documented. For example, among the Baka groups of southern Cameroon, wild yams are harvested in ways that actively promote plant regeneration, allowing for their sustained and recurrent exploitation (Dounias, 2001). Nevertheless, even acknowledging the potential existence of such low-level management practices—often archaeologically subtle and rarely preserved—we currently lack palaeoecological reconstructions, including detailed vegetation and fire histories, that would allow us to evaluate whether small-scale cultivation patches or garden-like spaces were prepared or maintained. Moreover, the majority of the rhizomatous taxa identified in the assemblage, such as *Heliconia*, *Monotagma*, and *Phenakospermum*, correspond to non-domesticated species, suggesting that their exploitation most likely involved the selective use and processing of wild geophytes rather than formally cultivated plants.

Our data indicate that subsistence practices at SLL were broad and generalised from the very outset of human occupation, exploiting a diversity of habitats within this ecotonal landscape, and centred on the use of palms, roots, and tree fruits.

The plant evidence is complemented by the faunal assemblage, which provides additional insight into the breadth of subsistence strategies practised by early foragers at SLL. Faunal remains, recovered only from Cerro Azul, Unit 1, show a predominance of fish, small and medium-sized mammals, reptiles, and trace amounts of bivalves (Morcote-Ríos 2021). The potential megafauna depicted in the rock art panels (Iriarte et al., 2022b) do not appear in the faunal record, and large mammals—abundantly represented in the rock art—are only minimally represented in the archaeological assemblage (Robinson et al., 2024). These patterns should nevertheless be interpreted with caution, given the high degree of fragmentation affecting the faunal remains (Osborne et al. in prep).

Taken together, the combined plant and faunal evidence indicates that the first territorial occupation of SLL was characterised by diversified and flexible subsistence strategies rather than by narrow or transitional subsistence regimes. From the beginning of human presence in the region, subsistence practices encompassed a wide spectrum of plant and animal resources, suggesting that early groups entered and occupied this landscape with well-developed, generalised economies already in place, structured around the local ecological mosaic of this transitional region between Amazonia and Orinoquia.

Our results align with a growing body of archaeobotanical and zooarchaeological research from diverse environmental settings worldwide, including temperate forests (Dillehay and Rossen, 2002), African forests (e.g., Mercader, 2002), the Near East (e.g., Zeder, 2012), Australia (Florin et al., 2020), and other tropical regions of the Americas (e.g., Robinson et al., 2021; Roosevelt et al. 1996; Shock and Moraes, 2019). Across these regions, early human groups frequently practised broad, flexible, and generalised subsistence strategies from the outset of occupation.

Another notable aspect of SLL is the long-term stability in plant use evident in the archaeobotanical record. From the earliest occupations onward, there is a consistent reliance on wild palms, roots, and tree fruits, indicating subsistence strategies that were highly resilient and well adapted to tropical forest environments. The appearance of domesticated plants, such as maize and manioc, is confined to the Late Holocene and represents a relatively recent and rather abrupt development within a long-standing pattern of foraging and forest resource management, rather than a fundamental economic transformation.

Turning to the mid-Holocene, a key point of discussion at the CM 1 archaeological site is the temporal interruption (hiatus) observed in both Units 2 and 3. In Unit 2, this hiatus spans between Levels 18 (8.34 cal ka BP) and 16 (1.87 cal ka BP).

Diachronic analysis of the archaeobotanical assemblage shows pronounced peaks in the abundance of most taxa—*Astrocaryum chambira*, *Attalea maripa*, *A. insignis*, *Bactris* sp., *Mauritia flexuosa*/*M. carana*, and *Brosimum lactescens*—relative to earlier levels. This pattern contrasts with the charcoal record, which indicates continuity in volume between the lower (Levels 18–21) and upper (Levels 15–12) strata (Figure 6). Neither the macrobotanical nor the phytolith evidence, however, provides a definitive explanation for the temporal hiatus observed at CM 1.

The available evidence indicates that the groups who reoccupied the rock shelters—particularly CM1—after the mid-Holocene hiatus continued to exploit broadly similar forest resources to those used during earlier occupations, but now within a subsistence system that incorporated plant cultivation. Following the hiatus, the ceramic-bearing levels show a marked increase in the abundance of macrobotanical remains, with pronounced peaks in most identified taxa (*Astrocaryum chambira*, *Attalea maripa*, *A. insignis*, *Bactris* sp., *Mauritia flexuosa*/*M. carana*, and *Brosimum lactescens*) compared with earlier levels. Although changes in charcoal frequency are less pronounced, total charcoal mass also tends to increase in the post-hiatus occupations (Figure 6).

This pattern can be explained by a combination of factors. On the one hand, ceramic-using populations likely intensified activities within the rock shelter, facilitating more efficient processing, preparation, and consumption of palm fruits. On the other, the greater abundance of macrobotanical remains may partly reflect better preservation of upper stratigraphic levels, whereas deeper deposits were more affected by compaction and post-depositional processes (Chrzazvez et al., 2014).

The shift in shelter use after the hiatus is consistent with evidence from nearby terra preta sites located at the base of the tepuis (e.g., Limoncillos rockshelter and Buenavista open air sites), dated to the first millennium BCE (Kosztura Nuñez and Morcote-Ríos, 2023). These sites contain prepared soils rich in organic matter and charcoal, together with maize phytoliths and carbonised seeds of the same palm taxa documented in the preceramic rock shelters, including CM1. Such soils have been interpreted as the result of fire-mediated landscape preparation for cultivation, aligning closely with the evidence for plant cultivation observed in the ceramic layers at CM1.

Together, these data suggest that the emergence of cultivable soils in the lower tepui landscapes from the first millennium BCE onward was associated with a new settlement pattern and a reconfiguration of rock-shelter use following the mid-Holocene hiatus. Although forest foraging practices persisted, as in the Early and Middle Holocene, they were now complemented by the cultivation of domesticated plants, including manioc and maize. In this sense, later populations reorganised their subsistence and settlement strategies around food production while maintaining long-standing modes of forest management.

The identification of Amazonian Dark Earths associated with the cultivation of domesticated plants represents the most significant economic shift documented in the region. Current evidence, however, is still insufficient to determine whether the ceramic occupations in the SLL reflect the arrival of new populations during the early Late Holocene. Although Late Holocene groups introduced ceramics and practised modified forms of landscape management, they maintained substantial continuity with earlier traditions, both in their use of forest resources and in cultural expressions such as rock art (Morcote-Rios et al. 2021).

Unlike other parts of Amazonia, the SLL shows no evidence for large-scale landscape modification, such as the construction of water-management canals, raised-field agriculture, or extensive burning, nor for forms of settlement nucleation or urbanism associated with intensive agricultural systems (Iriarte, 2024). These patterns suggest that, although domesticated crops were incorporated into local subsistence strategies, land use remained based on low-level food production and was broadly continuous with that of earlier populations.

Macrobotanical evidence from the CM archaeological site, related to pre-Columbian agriculture and domesticated plants in the SLL, reveals a small assemblage of domesticated, cultivated, and adventitious plants associated with ancient farming practices. This evidence outlines an important scenario in which three plant species appear directly linked to pre-Columbian agricultural systems: *Spondias mombin* (Anacardiaceae), *Phytolacca rivinoides* (Phytolaccaceae), and *Zea mays* (Poaceae).

Radiocarbon dates associated with these taxa indicate that by c. 2.7 cal ka BP (Late Holocene), agricultural practices were already established in the SLL. This chronology closely matches that of the *Terra Preta* (pre-Columbian agricultural soils) recorded in the same region, with an age of around 2.8 cal ka BP (Kosztura Nuñez and Morcote-Ríos, 2023). Two points stand out regarding domesticated species at CM. One is that the presence of maize extends only to the last 520 years, although the earliest records of this crop in the SLL date to c. 2.4 cal ka BP at the Buena Vista archaeological site. The other is that, strikingly, the peach palm (*Bactris gasipaes* var. *gasipaes*), a species domesticated in the Americas and of major cultural and dietary importance to many Indigenous Amazonian peoples, is absent from the SLL archaeological record. The remains identified as *Bactris* sp. at CM do not exhibit the diagnostic morphological traits of *B. gasipaes*, suggesting that this palm was likely introduced to the north-western Amazon only in more recent times.

Finally, we address an important topic that has received little attention in Amazonian archaeobotanical studies: medicinal plants. Their knowledge and use played a fundamental role in the lives of ancient human societies inhabiting the Amazonian tropical rainforest. Several plants identified at CM were not only key food resources but also likely served medicinal purposes: *Protium* sp., whose resin is used to treat colds (a disease introduced to the Americas by Europeans) and stomach ailments; *Hymenaea courbaril*, whose resin has medicinal applications; *Costus cf. spiralis*, used in infusions for liver conditions and kidney stones; *Phytolacca rivinoides*, employed in poultices and as a cicatrizant; and *Byrsonima spicata*, whose bark is medicinal (see more details in Supplementary Table SI5). The earliest evidence for medicinal plant use dates to around 7.8 cal ka BP, while the greatest diversity of medicinal species appears from c. 2.0 cal ka BP (Late Holocene) onwards.

Along with these plants of economic importance, one of the most intriguing findings at CM1 is the presence of saprotrophic fungi, which feed on decomposing organic matter. Their occurrence suggests that fuel procurement practices may have involved the collection of branches and logs already lying on the forest floor and undergoing decomposition. Such woody material could have hosted these fungi and, when gathered for combustion, would have been transported together with them to the settlement.

Archaeobotanical evidence from the SLL indicates the presence of tropical forest vegetation during the earliest human occupations. Thousands of carbonised seeds recovered from secure archaeological contexts are dominated by arboreal taxa characteristic of terra firme Amazonian forests, particularly palms (Palmae) and fruit-bearing trees. Identified palm taxa include *Astrocaryum chambira*, *Attalea maripa*, *A. racemosa*, *Bactris* sp., *Euterpe precatoria*, *Mauritia flexuosa*, *Oenocarpus bataua*, *O. minor*, and *Syagrus orinocensis*. In addition, seeds of *Brosimum lactescens* (Moraceae)—a dominant tree species of terra firme Amazonian forests—are present throughout the occupational sequence, indicating the sustained availability of forest resources in the immediate vicinity of the site.

At present, however, there is no local palaeoecological record from SLL itself. The closest available pollen sequences show contrasting environmental histories: the Loma Linda record to the north indicates the persistence of savannah landscapes since the Early Holocene (Behling and Hooghiemstra, 2000), whereas the Pantano de Mónica record (Behling et al., 1999) to the south documents the presence of tropical rainforest since the terminal Pleistocene. In this context, while the archaeobotanical assemblages from SLL strongly suggest a forested setting, targeted palaeoecological research is required to independently test and refine this interpretation and to better characterise the structure and extent of past vegetation at the local scale.

In sum, the multiproxy archaeobotanical record from the Serranía de La Lindosa demonstrates that early human groups

entering north-western Amazonia did so with well-developed, flexible subsistence systems structured around the reliable exploitation of palms, roots, and forest fruits along with fish and small and medium mammals.

Importantly, this study provides one of the first systematic archaeobotanical documentations of root and rhizome exploitation in the Colombian Amazon, made possible through the integration of phytolith, starch-grain, and charred parenchyma analyses. From the Late Pleistocene onward, plant use at CM1 reflects a stable and resilient adaptation to tropical forest environments, characterised by broad-spectrum foraging rather than early domestication or large-scale landscape transformation. The incorporation of domesticated crops occurs during the Late Holocene. By integrating macroremains, microfossils, faunal evidence, and regional comparisons, this study contributes not only to our understanding of long-term plant-use trajectories in the Colombian Amazon but also to broader debates on tropical forest colonisation, the emergence of cultivation, and the enduring role of diversified foraging economies in shaping Amazonian landscapes over millennial timescales.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

GM-R: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – original draft. FA: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – original draft. MR: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – original draft. SH: Formal analysis, Investigation, Methodology, Writing – original draft. JI: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. As none of the authors are native English speakers, generative AI tools were used in selected sections to improve the clarity of the language. All outputs were carefully reviewed and edited by the authors to ensure accuracy and intended meaning.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fearc.2026.1814264/full#supplementary-material

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